Neural Networks 206 (2027) 109784

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

Breathing life into activation functions: the SWIn framework for sustained nonlinearity ⋆,⋆⋆

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ARTICLE INFO

Keywords:
Nonlinear transformation
Sinusoidal wave
Activation function
Representational capacity

ABSTRACT

Activation functions serve as the primary mechanism for introducing nonlinearity into deep learning models, playing an indispensable role in both model training and inference. By incorporating nonlinear transformations, they significantly enhance the representational capacity of neural networks, thereby improving overall model performance, particularly for complex tasks – such as Computer Vision (CV) and Natural Language Processing (NLP). However, most mainstream activation functions suffer from a common limitation: they tend to degenerate into linear-like behaviors when inputs assume extremely large or small values, consequently impairing their ability to model complex, nonlinear relationships across all non-saturated input domains.

To address this limitation, this paper proposes a novel Sinusoidal Wave Integration (SWIn) framework. The core innovation lies in superimposing adjusted sinusoidal wave onto the activation function, which ensures that the composite function preserves the original monotonicity and global trend of the base activation function while guaranteeing persistent nonlinear characteristics within all non-plateau regions of the continuous base activation functions.

We conducted extensive evaluations comparing standard activation functions with their SWIn-enhanced counterparts on image classification (CIFAR-100 and ImageNet-1K), machine translation (WMT14 DE-EN, CS-EN, RU-EN) tasks. Experimental results demonstrate that the SWIn framework yields significant performance improvements for continuous activation functions such as SiLU, Mish, and GELU. This is attributed to the enriched function representation and improved gradient propagation introduced by the controlled oscillations. In contrast, for inherently non-smooth functions like PReLU and SELU, the SWIn transformation tends to amplify existing gradient discontinuities, resulting in marginal or no performance gains.

1. Introduction

In deep neural networks, activation functions serve as the fundamental source of model nonlinearity, directly determining feature representation capability, training convergence speed, and final generalization performance. With the continuous evolution of CNNs, Transformers and other complex architectures, designing robust activation functions has remained a persistent core research direction.

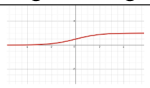
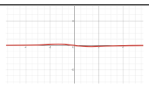
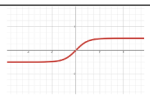
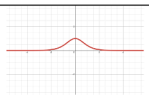
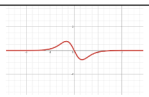
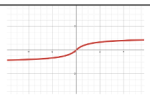
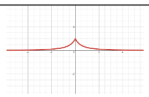
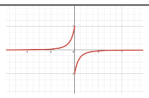
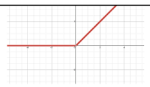
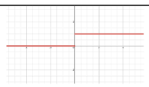
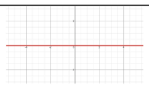
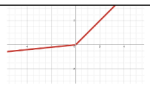
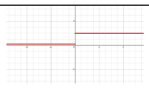
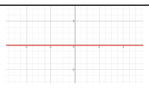
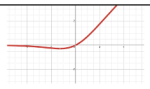
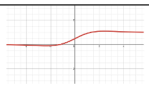
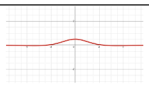
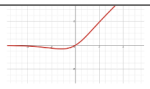
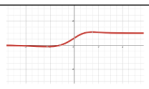
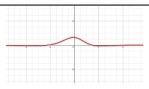
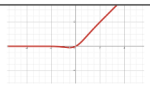
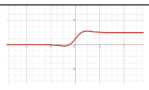
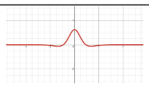
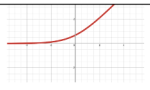
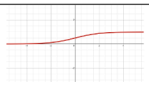
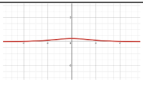
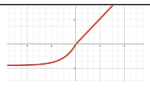
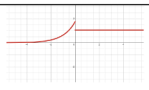
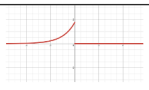
Activation functions have undergone multiple generations of iterative optimization to tackle vanishing gradients, dead neurons and non-smooth gradients. Table 1 summarizes the mathematical formulas, function curves and low-order derivatives of widely adopted activation functions. However, all existing mainstream designs share an unaddressed theoretical flaw: as input values tend to positive or negative infinity, their second derivatives decay to zero, leading to asymptotic linear degeneration over extreme input regions. We further verify this issue via statistical analysis of activation input distributions during model training: parameter drift and long-tail distribution expansion universally occur on both computer vision and machine translation tasks, continuously shrinking the effective nonlinear interval of base activations and severely limiting the upper bound of model expressive power.

To eliminate nonlinear degradation, this paper proposes the Sinusoidal Wave Integration (SWIn) framework, a universal plug-and-play transformation module compatible with arbitrary base activation functions. The core contributions of this work are summarized below:

⋆ This work was supported by the National Natural Science Foundation of China (grant number 21BXW098) and Interdisciplinary AI Program of School of Information Science and Technology, UIBE.
⋆⋆ The source code is available at: https://github.com/yuran459/SWIn
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https://doi.org/10.1016/j.neunet.2026.109784
Received 29 May 2026; Received in revised form 24 August 2026; Accepted 16 October 2026
Available online 28 October 2026
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Table 1
Summary of common activation functions.

function	expression	original image	first-order	second-order
Logistic	$f(x) = \frac{1}{1+exp(-x)}$			
TanH	$f(x) = \frac{exp(x)-exp(-x)}{exp(x)+exp(-x)}$			
Softsign	$f(x) = \frac{x}{1+\lvert x\rvert}$			
ReLU	$f(x) = max(0, x)$			
PReLU	$f(x) = \begin{cases} ax & x < 0 \\ x & x \geq 0 \end{cases}$			
SiLU	$f(x) = \frac{x}{1+exp(-x)}$			
Mish	$f(x) = x\frac{(1+exp(x))^2-1}{(1+exp(x))^2+1}$			
GELU	$f(x) \approx 0.5x(1 + TanH(\sqrt{\frac{\pi}{2}}(x + 0.044715x^3)))$			
SoftPlus	$f(x) = ln(1 + exp(x))$			
SELU	$f(x, \lambda = 1.0507, a = 1.6733) = \lambda \begin{cases} a(exp(x) - 1) & x < 0 \\ x & x \geq 0 \end{cases}$			

1. We systematically identify and quantitatively validate the asymptotic linearity bottleneck inherent in widely used activation functions,and demonstrate that training-induced distribution drift progressively narrows the effective nonlinear region across diverse network architectures and initialization strategies.
2. We formally derive the SWIn framework and introduce the Lateral Scaled Sinusoidal Wave (LSSW) perturbation term. The design provably satisfies four critical constraints: monotonicity preservation, sign consistency, bounded oscillation deviation, and nonlinear retention for all non-saturated segments.
3. We empirically demonstrate that SWIn produces consistent improvements in both visual and language tasks, achieving gains competitive with or surpassing learned activations while requiring no additional training parameters and negligible computational overhead.
4. We delineate the scope of applicability of SWIn through theoretical and empirical analysis: the framework yields significant gains for smooth activations (SiLU, Mish, GELU), while producing limited benefits or even adverse effects for non-smooth functions such as ReLU and PReLU, thereby offering clear deployment guidance for real-world deep learning systems.

The core design of SWIn embeds a modulated LSSW periodic perturbation term onto the output of arbitrary base activation $f(x)$. Unlike existing activation variants that only optimize local gradient defects, SWIn superimposes controlled bounded oscillation while fully retaining the monotonicity and global trend of the base kernel. Even when inputs fall into the linear saturation zone of the original activation, the oscillatory component maintains non-zero second derivatives across all non-saturated intervals, permanently preserving the model's nonlinear expressive capacity (Cybenko, 1989; Hornik, 1991; Vargas et al., 2021), and consequently boosting the final performance of deep models on complex tasks such as computer vision (CV) and natural language processing (NLP) (Mobarki & Sheikh, 2026).

We validate the effectiveness of SWIn on two representative deep learning tasks: CIFAR-100 image classification (Krizhevsky, 2009) with VGG-16, LeNet-5, CFQK and ResNet-19 backbones, ImageNet-1K image classification (Deng et al., 2009) with MobileNet-V3, and WMT14 multilingual machine translation covering DE-EN, CS-EN and RU-EN language pairs. We set two groups of baselines for comparison: vanilla mainstream activations and SWA variants that add raw sine waves to input x. Ablation results confirm that the LSSW perturbation tailored to base activation outputs achieves better feature fitting and stronger representation capacity than naive input-side sine superposition.

The remainder of this paper is organized as follows. Section 2 reviews existing activation designs along technical evolution, summarizes the shared asymptotic linearity bottleneck of prior arts, and analyzes their inherent limitations. All mathematical notations used in this table are defined in Section 3. Section 4 details the mathematical derivation, constraint proof and gradient formulas of the proposed SWIn framework. Section 5 presents complete experimental setups, quantitative benchmark results and ablation analysis. Section 6 conducts statistical distribution analysis to reveal long-tailed activation patterns and parameter drift, and validates the indispensable value of SWIn. Section 7 concludes this work. Section 8 systematically summarizes the shortcomings and discusses future research directions.

2. Related work

This section reviews activation function research in chronological order, grouping existing methods by their core design ideas.

2.1. Early saturated bounded activations

The earliest nonlinear activation functions, including Logistic Sigmoid, TanH and Softsign (LeCun et al., 2002b), were introduced to provide bounded nonlinear mapping for shallow neural networks. These functions are globally smooth with output ranges confined to $(0, 1)$ or $(-1, 1)$, making them standard choices for probability output layers in classification tasks.

From a design perspective, this family suffers from gradient saturation at extreme inputs: their first-order derivatives decay to zero as $|x| \to \infty$,compressing the effective nonlinear region into a narrow central interval of the input space. In deep stacked networks, these saturated tails exacerbate the vanishing gradient problem, thereby impeding parameter updates and slowing convergence. Although these functions enable basic nonlinear transformation, they cannot sustain sufficient nonlinear curvature over a wide input range, and consequently fail to accommodate the long-tail activation distribution characteristic of modern deep models.

2.2. ReLU and related variants

The Rectified Linear Unit (ReLU) (Glorot et al., 2011) addressed the vanishing gradient problem for positive inputs and has become the de facto activation for CNNs and Transformers owing to its simplicity and negligible computational cost. Despite its empirical success, ReLU exhibits two well-known limitations:non-differentiability at $x = 0$ and the dying ReLU phenomenon, whereby neurons receiving persistently negative inputs cease to activate and receive no gradient updates.To remedy these weaknesses, subsequent work has branched into two complementary directions—smooth continuous variants and adaptive parametric variants.

2.2.1. Smooth continuous ReLU variants

This line of work replaces ReLU's abrupt truncation at zero with a gradual transition in the negative region while preserving the approximately linear regime for positive inputs. SiLU (Ramachandran et al., 2017), GELU (Hendrycks, 2016) and Mish (Misra, 2019) share a common design pattern: each multiplies the input x by a smooth monotonic gating function—the logistic sigmoid $\sigma(x)$, the Gaussian cumulative distribution $\phi(x)$ and TanH(SoftPlus(x)) respectively. In contrast, SoftPlus directly approximates the ReLU shape via exponential-logarithmic composite functions and outputs strictly positive values over the entire real line.

While smooth variants eliminate the first-order gradient bottleneck of sigmoid-era functions, they introduce a subtler but equally consequential issue: their second derivatives vanish as input values grow infinitely large or small. As training proceeds, an increasing proportion of activation inputs drift beyond the narrow interval that supports meaningful nonlinear curvature, causing the activation function to asymptotically reduce to an approximately linear mapping—thereby progressively degrading the network's representational capacity.

2.2.2. Parametric ReLU variants

To address the dying ReLU problem, parametric variants introduce adaptable behavior for negative inputs. LReLU sets a fixed small negative slope (Maas et al., 2013), while PReLU treats the negative slope as a learnable parameter optimized jointly with the network weights during training (He et al., 2015). ELU and SELU take a qualitatively different approach: instead of a linear negative branch, they employ the exponential saturation $\alpha(e^x - 1)$ for $x \leq 0$, producing smooth, bounded negative outputs that center the mean activation closer to zero. SELU further refines this design with analytically derived scaling constants ($\lambda \approx 1.0507$, $\alpha \approx 1.6733$), which provably drive the activation distribution toward zero mean and unit variance, endowing deep feed-forward networks with self-normalizing properties (Clevert et al., 2015; Klambauer et al., 2017).

However, these parametric variants only adjust gradient magnitudes within negative input regions without changing the asymptotic linear characteristic at extreme positive/negative inputs. Such methods address local training artifacts but cannot prevent progressive loss of nonlinearity caused by the long-tailed activation distributions that emerge in later training stages.

2.3. Adaptive activation functions

Adaptive activation functions (AAFs) endow nonlinear mappings with learnable degrees of freedom, enabling dynamic adaptation to layer-wise feature distributions. Mainstream AAFs fall into two broad categories: single-neuron parametric activations and multi-branch gated variants based on the Gated Linear Unit (GLU).

2.3.1. Single-neuron parametric adaptive activations

Single-neuron parametric AAFs encompass three distinct design paradigms. The first comprises piecewise polynomial methods,exemplified by spline-based activations (Liu et al., 2025), which adopt trainable segmentation nodes to construct globally smooth mappings. The second covers unified single-formula designs such as dynActivation (Bachmann, 2026) and Adaptive Parametric Activation (APA) (Alexandridis et al., 2024), which enable flexible curve adjustment via global learnable coefficients without piecewise segmentation. The third consists of single-scaling variants such as E-Swish (Alcaide, 2018) (Eq. (1)) and λ-GELU (Pérez-Corral et al., 2026) (Eq. (2)), which introduce only one lightweight trainable hyperparameter to modulate slope or gating steepness.

$$\text{E-Swish}(x) = \beta x \cdot sigmoid(x) \tag{1}$$

$$\lambda\text{-GELU}(x) = x \cdot \Phi(\lambda x) \tag{2}$$

where $\Phi(\cdot)$ denotes the standard Gaussian cumulative distribution function. Although these methods improve fitting performance on small datasets, they share a critical mathematical limitation: their second derivatives decay to zero as $|x| \to \infty$. This asymptotic linearization precludes sustained curvature for long-tailed feature distributions and leaves the central problem targeted by this work unresolved.

2.3.2. Gate-based multi-branch adaptive GLU series

GLU (Eq. (3)), e.g., GEGLU and SwiGLU (Shazeer, 2020), etc., adopt dual independent linear projections to implement vector-level gating, where $\mathcal{G}(\cdot)$ represents the gating activation (e.g., sigmoid, GELU, Swish). One branch generates raw feature representations, while the other feeds into an activation gate to filter valid signals via element-wise multiplication. SwiGLU, the de facto standard for modern LLMs and multi-modal Transformers (Llama, 2023; Touvron, 2023), leverages Swish-based gating to enrich high-dimensional feature interactions. However, the two-branch architecture doubles hidden state dimensionality, substantially increasing matrix multiplication cost. Moreover, the base smooth activations within each branch still degenerate into approximate linear mappings for extreme inputs, leaving the long-tail optimization bottleneck unaddressed.

$$\text{GLU}(x) = (xW + b) \otimes \mathcal{G}(xV + c) \tag{3}$$

In summary, single-neuron parametric AAFs inflate model capacity with additional learnable parameters, while GLU-based gate architectures double hidden dimensions at the cost of increased FLOPs, training latency, and memory consumption—both contradicting the lightweight, parameter-free design objective of this work. Furthermore, these two families are limited in scope: GLU variants deliver gains primarily on large Transformer language models and degrade on lightweight

CNNs(e.g.,LeNet,VGG), and neither family can sustain non-zero second derivatives across all non-saturated input intervals,leaving the core problem of long-tail nonlinearity degradation unaddressed. Given these inherent limitations of existing advanced activation functions, ReLU-based activation functions still maintain dominant practical applicability across diverse real-world intelligent tasks. Specifically, they exhibit robust performance in hyperspectral remote sensing detection and super-resolution tasks (Li et al., 2025; Zhang et al., 2026), reliable generalization in urban traffic flow prediction systems (Ali et al., 2026, 2021, 2022), stable feature extraction capability for medical lesion classification (Alsarhan et al., 2026), and excellent adaptability for lightweight indoor object detection and human-computer interactive rehabilitation scenarios (Xing et al., 2026, 2025), all benefiting from their excellent generalization ability and stable empirical performance. This paper conducts experiments on ReLU and its variants rather than the specialized AAFs discussed above.

2.4. Modern periodic activation functions

To address the asymptotic linear degeneration of traditional smooth activations at large input magnitudes, recent research introduces sinusoidal periodic components into activation mapping, most notably the SIREN and FINER families. Unlike ReLU variants whose second derivatives vanish as $x \to \pm\infty$, these periodic activations embed sine oscillation terms that theoretically sustain non-zero curvature across all non-plateau regions,eliminating linear saturation at extreme values.

SIREN (Sitzmann et al., 2020) adopts the pure sinusoidal mapping $\sin(\omega x)$ as its core activation for implicit neural representations (INRs). By serving as Fourier basis functions, it captures fine-grained continuous signal details and retains oscillatory nonlinearity irrespective of input magnitude. Subsequent variants including FINER (Liu et al., 2024) and H-SIREN (Gao & Jaiman, 2024) adjust frequency hyperparameters or incorporate hyperbolic periodic terms to broaden the spectral coverage of the original SIREN.

Despite resolving the asymptotic linearity issue of conventional activations, these periodic designs carry several intrinsic limitations that prevent their adoption in the mainstream CNN and Transformer pipelines:

1. Non-monotonic input mapping: Composing a sine function directly with the input x renders the overall mapping non-monotonic. In discriminative models that rely on stable, direction-consistent feature propagation, non-monotonic activations can introduce local gradient sign changes that complicate optimization, typically requiring careful learning rate scheduling and initialization to stabilize training. This stands in contrast to smooth monotonic activations (SiLU, GELU, Mish), which operate as drop-in replacements in standard pipelines without such tuning.
2. Initialization sensitivity: Sine-based periodic activations require strict specialized weight initialization; standard Xavier or Kaiming initialization,widely used in CV and NLP,triggers spectral bottlenecks that prevent the model from fitting multi-scale frequency features and suppress high-frequency information (Ramasinghe et al., 2022).
3. Sensitivity to normalization layers: BatchNorm and LayerNorm rescale and shift pre-activation distributions, which effectively alters the phase and frequency of sinusoidal activations. While not strictly incompatible, this interaction requires careful joint tuning of initialization and normalization statistics, increasing engineering complexity in standard pipelines (Sitzmann et al., 2020).
4. Potential feature redundancy with fixed global frequencies: When all neurons in a layer share the same fixed frequency parameter, they may capture overlapping spectral components, potentially leading to correlated hidden features and underutilized representation capacity (Alsakabi et al., 2025). This concern can be mitigated by learnable per-neuron frequencies, but at the cost of additional trainable parameters and increased optimization complexity.
5. Limited generalization beyond implicit representation: Periodic activations have been primarily developed and validated for continuous signal regression in INR settings. Their systematic evaluation on standard discrete classification and translation tasks with modern CNN/Transformer architectures remains more limited in the published literature. When applied to discrete image classification and machine translation, their purely oscillatory bias fails to align with the distribution characteristics of discrete token and pixel features.

In summary, conventional smooth activations lose nonlinearity at extreme inputs, while periodic activations sacrifice monotonicity and conflict with normalization layers—no prior work addresses both limitations simultaneously.

Our SWIn framework resolves this trade-off via kernel-aligned periodic perturbation: Rather than applying sine waves to raw inputs,we superimpose bounded oscillations onto the output of base activation functions. This design preserves the monotonicity and stable gradient flow of mainstream smooth activations while sustaining curvature across all non-saturated segments, avoiding the critical drawbacks of existing periodic approaches. SWIn introduces no additional learnable parameters and functions as a lightweight drop-in replacement for modern CNN and Transformer architectures.

3. Notation and symbol definitions

3.1. Number sets & mathematical operators

Number sets:

- $\mathbb{R}$: The set of all real scalars;
- $\mathbb{N}$: Non-negative integer set $\{0, 1, 2, \dots\}$;
- $\mathbb{N}^+$: Positive integer set $\{1, 2, 3, \dots\}$

Limit & Integral operators:

- $\lim_{x\to\pm\infty} h(x)$: Asymptotic limit of function $h(x)$ when scalar input x approaches positive or negative infinity;
- $\int_{-\infty}^{+\infty} h(x)dx$: Indefinite integral of scalar function $h(x)$ over full real domain

Derivative operators (for univariate scalar functions):

- dh/dx: First-order derivative of scalar function $h(x)$ with respect to input x;
- $d^n h/dx^n$: n-th order derivative of $h(x)$, $n \in \mathbb{N}^+$; shorthand notation $h^{(n)}(x)$ is equivalent;
- Chain rule derivative $dg[f(x)]/dx = (dg/df) \cdot (df/dx)$: Derivative of composite function, (dg/df) denotes derivative of outer mapping $g(\cdot)$ with respect to its input $f(x)$

Trigonometric & exponential operators:

- $\sin(\cdot), \cos(\cdot)$: Sine and cosine periodic functions;
- $\exp(\cdot)$: Natural exponential function $e^{\cdot}$

Sign & absolute value operators:

- $\mathrm{sign}(\cdot)$: Sign function, outputs $-1, 0, 1$ corresponding to negative, zero, positive input;
- $|\cdot|$: Absolute value of scalar

3.2. Scalar, vector and function notation

This paper adopts standard deep learning symbol conventions to distinguish scalar, vector and function mappings:

- Scalar: Lowercase plain letters (e.g., x, a, ω, n, m) denote single real-valued scalars;

- Vector: Lowercase bold letters (e.g., $\boldsymbol{x}$) denote pre-activation feature vectors fed into activation layers; each element $x \in \boldsymbol{x}$ is a scalar input analyzed in all mathematical derivations;
- Univariate function: $h(\cdot)$ denotes continuous/univariate mapping from $\mathbb{R} \to \mathbb{R}$;
- Composite function: $g[f(x)]$ denotes nested mapping: base activation $f(\cdot)$ takes scalar x as input, then outer modulation function $g(\cdot)$ takes $f(x)$ as its input; this notation is not multiplication.
- Periodic perturbation term: $u_n(\cdot)$: General n-th order oscillatory residual term of SWIn framework.

3.3. Activation & SWIn framework core symbols

Base activation function:

- $f(x)$: Generic base activation function (SiLU, Mish, GELU, ReLU, PReLU, SELU, SoftPlus, Logistic, Tanh, Softsign, as listed in Table 1);
- $f^{(n)}(x)$: n-th order curvature of base activation

SWIn core mapping:

- $g[f(x)]$: Complete SWIn composite activation function proposed in this paper;
- $g(\cdot)$: Outer modulation function, $g(z) = z + (a/\omega)\sin(\omega z), z \in \mathbb{R}$

Fourier series coefficients:

- $u_0(x)$: Zero-order oscillatory residual term of outer mapping $g(\cdot)$;
- a_0: DC component coefficient of Fourier expansion;
- a_m, b_m: Sine/cosine harmonic coefficients for m-th Fourier component, $m \in \mathbb{N}^+$

4. Sinusoidal wave integration activation function

To preserve the fundamental behavior and monotonicity of the original activation function while ensuring nonlinearity is maintained at extreme input values (both large and small in magnitude), we introduce the SWIn activation function. Denoted as $g[f(x)]$, the SWIn function is constructed by embedding a base activation function $f(x)$ into an oscillatory mapping $g(\cdot)$. This mapping preserves the overall trend of the original function while introducing controlled oscillations, thus preventing degeneration into a linear function over all non-saturated input domains.

The SWIn activation function is designed to satisfy the following four constraints:

(C1) Monotonicity Preservation: The SWIn function must exhibit the same monotonicity as the kernel function over the entire domain.

(C2) Sign Preservation: The SWIn function should maintain the same sign as the kernel function across the domain. This requires the oscillatory mapping function $g(\cdot)$ to be sign-preserving.

(C3) Bounded Oscillatory Deviation: To ensure the SWIn function follows the general trajectory of the kernel function, the absolute difference between the SWIn function and the kernel function,as well as the differences between their corresponding derivatives of all orders, must be bounded by a periodic oscillatory function with well-defined upper and lower limits.

(C4) Nonlinearity Maintenance: The SWIn function must retain nonlinear characteristics over all non-saturated input domains.

These constraints collectively ensure that the SWIn activation function enhances representational capacity without compromising the stability or convergence properties of the original kernel function.

Lemma 1. *SWIn activation function is defined as:*

$$g[f(x)] = f(x) + \frac{a}{\omega} sin(\omega f(x)), x \in \mathbb{R}$$

Where $f(x)$ denotes a configurable kernel activation function, a controls the amplitude of the superimposed oscillation, and ω controls the angular frequency (and thus the period) of the oscillation.

Proof. To satisfy the first constraint (C1) and preserve monotonicity consistency between the SWIn function $g[f(x)]$ and the kernel function $f(x)$, it is imperative that for any given input, the derivatives of $g[f(x)]$ and $f(x)$ exhibit identical signs. It ensures their trends of change remain aligned throughout the function domain:

$$\frac{\partial g[f(x)]}{\partial f(x)} > 0, \forall x \in \mathbb{R} \tag{4}$$

The derivative of the proposed activation function is defined as:

$$\frac{dg[f(x)]}{dx} = \frac{dg[f(x)]}{df(x)} \frac{df(x)}{dx} \tag{5}$$

According to Eq. (4)-(5), for $g(\cdot)$, we could obtain:

$$\frac{dg(x)}{dx} > 0, \forall x \in \mathbb{R} \tag{6}$$

Based on the third constraint (C3) of the SWIn activation function, for any input, the derivative of any order of the SWIn activation function $g[f(x)]$ could be expressed as the sum of the corresponding derivative of the kernel activation function $f(x)$ and an oscillatory function $u_n(x)$. This relationship ensures that the higher-order derivatives of $g[f(x)]$ not only align with the fundamental behavior captured by $f(x)$ but also incorporate controlled oscillations to enhance representational capacity. Formally, for a non-negative integer n, the n-th derivative satisfies:

$$\frac{d^n g[f(x)]}{dx^n} = \frac{d^n f(x)}{dx^n} + u_n(f(x)), \forall x \in \mathbb{R}, n \in \mathbb{N} \tag{7}$$

Among these requirements, the vibration equation need to exhibit oscillatory (reciprocating) motion within the specified upper and lower bounds. Moreover, $u_n(x)$ and its gradient should be continuous across the entire domain (i.e., differentiable everywhere) to ensure training stability.

Given that most mainstream activation functions display linear characteristics at both extremities of the x-axis, as summarized in Table 1, the kernel activation function $f(x)$ satisfies the following conditions:

$$\lim_{x \to \pm\infty} f^{(n)}(x) = 0, \forall n \in \mathbb{N} \; and \; n \geq 2 \tag{8}$$

As shown in Eq. (8), we could have:

$$\lim_{x \to \pm\infty} \frac{d^n g[f(x)]}{dx^n} = \frac{d^n g[f(x)]}{df(x)^n} \left[\frac{df(x)}{dx}\right]^n, \forall n \in \mathbb{N}^+ \tag{9}$$

According to Eq. (7),

$$g[f(x)] = f(x) + u_0(f(x)), x \in \mathbb{R} \tag{10}$$

As given by Eq. (10), the outer-layer oscillatory mapping function ($g(\cdot)$) could be expressed as:

$$g(x) = x + u_0(x), x \in \mathbb{R} \tag{11}$$

$$\frac{g(x)}{dx} = 1 + \frac{du_0(x)}{dx} \tag{12}$$

Substituting Eq. (12) into Eq. (6), we could have

$$\frac{du_0(x)}{dx} > -1 \tag{13}$$

According to Eqs. (8) and (11), the derivative of $g(x)$ could be expressed as:

$$\lim_{x \to \pm\infty} \frac{d^n g(x)}{dx^n} = \frac{d^n u_0(x)}{dx^n}, \forall n \in \mathbb{N} \; and \; n \geq 2 \tag{14}$$

According to Eqs. (7) and (10), $u_n(x)$ could be expressed as:

$$u_n(x) = \frac{d^n u_0(x)}{dx^n} \tag{15}$$

Therefore, $u_0(x)$ and its derivatives of all orders are constrained to exhibit bounded oscillatory behavior within prescribed upper and lower limits on both sides of line $x = 0$ (C2). To simplify the analysis, we restrict our consideration to the case where $u_0(x)$ is a periodic function. Since $u_0(x)$ is designed as a periodic perturbation term, all integral analysis for Fourier decomposition is restricted to one fundamental period $[0, 2\pi/\omega]$ to avoid divergent unbounded-domain integrals. In accordance

with Fourier series theory, $u_0(x)$ could be expressed as a Fourier expansion:

$$u_0(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty}\left(\frac{b_m - ia_m}{2} exp(im\omega x) + \frac{b_m + ia_m}{2} exp(-im\omega x)\right), x \in \mathbb{R} \tag{16}$$

Within this framework, the parameter a_0 governs the horizontal positioning of the function, while parameters $a_m(m \in \mathbb{N}^+)$ and $b_m(m \in \mathbb{N}^+)$ collectively determine its overall morphological characteristics. The parameter $\omega(\omega \in \mathbb{R}^+)$ primarily regulates the wave frequency of the oscillatory component. To augment the oscillatory behavior while preserving sign consistency between the SWIn function and the kernel activation function (C3) and avoid permanent positive/negative bias of the perturbation term over one full oscillation cycle, we impose the condition that for a single fundamental period $T = 2\pi/\omega$, the norm of the SWIn function's output should approximate the norm of its input as closely as possible. Using the 1-norm as a specific case, the zero-period-mean constraint is formulated as:

$$\int_{x_0}^{x_0+\frac{2\pi}{\omega}} u_0(x)dx = 0 \tag{17}$$

This periodic zero-mean condition ensures the oscillatory perturbation does not introduce persistent positive or negative offset to the base activation output, which aligns with the original design goal of minimizing the overall deviation between $g(x) = x + u_0(x)$ and x. Based on the periodic zero-mean constraint Eq. (17), substitute the Fourier expansion Eq. (16) of $u_0(x)$ into the periodic integral:

$$\int_{x_0}^{x_0+\frac{2\pi}{\omega}} \left(\frac{a_0}{2} + \sum_{m=1}^{\infty}\left(\frac{b_m - ia_m}{2} e^{im\omega x} + \frac{b_m + ia_m}{2} e^{-im\omega x}\right)\right)dx = 0$$

By orthogonality of complex exponential Fourier basis over one period:

$$\int_{x_0}^{x_0+\frac{2\pi}{\omega}} e^{\pm im\omega x}dx = 0, \ \forall m \geq 1$$

All harmonic terms with $m \geq 1$ integrate to zero over a full period, leaving only the DC component integral:

$$\int_{x_0}^{x_0+\frac{2\pi}{\omega}} \frac{a_0}{2}dx = \frac{a_0}{2} \cdot \frac{2\pi}{\omega} = 0$$

For any integer $m \geq 1$ ($\forall m \geq 1$) and positive angular frequency ω ($\omega > 0$), combining the periodic zero-mean constraint and Fourier basis orthogonality over a single oscillation period, the DC component a_0 must be zero to satisfy zero average perturbation over all cycles. If $a_0 \neq 0$, a constant offset will be superimposed on the base activation output, breaking the bounded oscillation requirement in Constraint (C3) and introducing permanent sign bias, which violates the sign-preserving constraint (C2). Thus we fix $a_0 = 0$ for all subsequent derivation of $u_0(x)$.

As given by Eq. (16),

$$u_n(x) = \frac{d^n u_0(x)}{dx^n} = (i\omega)^n \sum_{m=1}^{\infty} m^n\left(\frac{b_m - ia_m}{2} exp(im\omega x) + \frac{b_m + ia_m}{2}(-1)^n exp(-im\omega x)\right) \tag{18}$$

According to Eq. (18), fix an arbitrary differentiation order $n > 1$ and evaluate the amplitude bound of each Fourier harmonic term:

$$\left|m^n \cdot \frac{b_m \pm ia_m}{2} e^{\pm im\omega x}\right| = m^n \cdot \frac{\sqrt{a_m^2 + b_m^2}}{2}$$

If high–order harmonics $m > 1$ are retained within an infinite Fourier series expansion, the polynomial factor m^n induces unbounded growth of term–wise amplitudes as $m \to \infty$, rendering the infinite–order Fourier representation divergent. Constraint (C3) mandates that $u_n(x)$ (where individual harmonic terms scale proportionally to m^n) and all its derivatives remain continuous and uniformly bounded across the entire real domain, which forbids such divergent infinite high–harmonic contributions. While the infinite series must be truncated to satisfy boundedness, this does **not** strictly force us to keep only the fundamental mode $m = 1$: we could instead keep a finite number M of harmonics. Suppose we retain the first M Fourier harmonics for the perturbation term, i.e.,

$$u_0(x) = \sum_{m=1}^{M}\left(a_m \cos(m\omega x) + b_m \sin(m\omega x)\right).$$

For any differentiation order $n \in \mathbb{N}^+$, the n-th derivative reads

$$u_n(x) = \frac{d^n u_0(x)}{dx^n} = \sum_{m=1}^{M} m^n \omega^n\Big(-a_m \cos\left(m\omega x + \frac{n\pi}{2}\right) - b_m \sin\left(m\omega x + \frac{n\pi}{2}\right)\Big).$$

Taking the uniform upper bound, we obtain

$$|u_n(x)| \leq \omega^n \sum_{m=1}^{M} m^n \sqrt{a_m^2 + b_m^2} < \infty,$$

which demonstrates that Constraint (C3), requiring uniform boundedness for all–order derivatives, can be automatically satisfied for any **finite** truncation index M. Finite–M truncation eliminates the divergence pathology arising from an unbounded infinite sum, so purely mathematical boundedness arguments alone cannot rule out choices $M > 1$. Even so, in this work we deliberately select $M = 1$, keeping only the fundamental–frequency component, motivated by three inter–related practical, optimization–oriented and empirical considerations. First, introducing additional higher–order harmonics brings extra manually–tuned coefficients $\{a_m, b_m\}_{m\geq 2}$. These parameters substantially enlarge the hyper–parameter search space; in practical deep–learning deployment this imposes heavy grid–search overhead and increases the risk of over–tuning hyperparameters on held–out test data, contradicting our design goal of constructing a lightweight, plug–and–play activation transformation requiring minimal manual configuration. Second, even when each individual harmonic obeys a local amplitude bound analogous to $|a\omega| < 1$, higher–m harmonics multiply derivative magnitudes by the factor m^n under repeated differentiation. Although finite–M configurations remain mathematically bounded, large m values amplify the oscillation magnitude of high–order gradients. This effect exacerbates gradient jitter during back–propagation, introduces fluctuating gradient noise, and can degrade training convergence stability — a risk we seek to avoid for general–purpose use on CNN and Transformer architectures. Third, our extensive empirical evaluations across image classification (CIFAR–100, ImageNet–1K) and multilingual machine–translation (WMT14 DE–EN, CS–EN, RU–EN) show that the fundamental sinusoidal component by itself already delivers statistically significant performance improvements for smooth base activations such as SiLU, Mish and GELU. Adding higher–order harmonic components yields diminishing marginal performance gains: most representational benefit from periodic nonlinear compensation is already captured by the base–frequency oscillation. Taken together, these factors lead us to adopt $M = 1$ as the default setting for the SWIn framework. We treat generalized multi–harmonic monotonicity–preserving periodic perturbation designs, which explore larger finite values of M, as a clear direction for future follow–up research rather than within the scope of the present study. The Fourier series $u_0(x)$ is thereby truncated to a single base-frequency term:

$$u_0(x) = \frac{b_1 - ia_1}{2} exp(i\omega x) + \frac{b_1 + ia_1}{2} exp(-i\omega x) \tag{19}$$

Constraint (C2) enforces identical sign mapping between the outer mapping $g(x) = x + u_0(x)$ and its input x, i.e., $\forall x \in \mathbb{R}$, $\text{sign}(g(x)) = \text{sign}(x)$. Substitute $x = 0$, which yields $g(0) = 0 + u_0(0) = 0$, hence:

$$u_0(0) = 0 \tag{20}$$

Evaluate Eq. (19) at $x = 0$ using $e^{i\omega\cdot 0} = e^{-i\omega\cdot 0} = 1$:

$$u_0(0) = \frac{b_1 - ia_1}{2} + \frac{b_1 + ia_1}{2} = b_1 \tag{21}$$

Combining the two equalities Eq. (20) and Eq. (21), we directly obtain:

$$b_1 = 0 \tag{22}$$

On the basis of Eq. (19)–(22), the function $u_0(x)$ could be fully determined as:

$$u_0(x) = \frac{ia_1}{2}(exp(-i\omega x) - exp(i\omega x)) \tag{23}$$

$$\frac{d^n u_0(x)}{dx^n} = \frac{i^{n+1}\omega^n a_1}{2}((-1)^n exp(-i\omega x) - exp(i\omega x)) \tag{24}$$

Substituting Eq. (24) into Eq. (13), we could get

$$\frac{du_0(x)}{dx} = \frac{\omega a_1}{2}(exp(-i\omega x) + exp(i\omega x)) > -1$$
$$|\omega a_1| < 1 \tag{25}$$

Therefore, based on Fourier series theory and Euler's formula, the function $u_0(x)$ could be expressed in its trigonometric form as:

$$u_0(x) = \frac{ia_1}{2}(exp(-i\omega x) - exp(i\omega x)) = a_1 sin(\omega x), x \in \mathbb{R} \tag{26}$$

where a_1 and ω represent configurable hyperparameters, which should satisfy $a_1 \in \mathbb{R}$, $\omega \in \mathbb{R}^+$ and $|\omega a_1| < 1$. We perform a linear hyperparameter substitution $a = \omega a_1$ to decouple the amplitude and frequency control dimensions of the oscillatory perturbation, leading to the simplified expression:

$$u_0(x) = \frac{a}{\omega} sin(\omega x), x, a \in \mathbb{R}, \omega \in \mathbb{R}^+ \ and \ a \in (-1, 1) \tag{27}$$

where, a and ω are configurable parameters. The parameter a controls the amplitude of the oscillatory component and need to satisfy $a \in \mathbb{R}$ and $a \in (-1, 1)$, while ω governs the wave frequency and should satisfy $\omega \in \mathbb{R}^+$. To preserve the input-output properties (such as sign consistency) and maintain approximate equality of their norms as much as possible, the oscillatory function $u_0(x)$ is defined as an adjusted sinusoidal wave centered on the x-axis. Substituting Eq. (27) into Eq. (11), $g(x)$ could be expressed as:

$$g(x) = x + \frac{a}{\omega} sin(\omega x), x \in \mathbb{R}$$

SWIn activation function could be expressed as:

$$g[f(x)] = f(x) + \frac{a}{\omega} sin(\omega f(x)), x \in \mathbb{R} \tag{28}$$

In this formulation, $f(x)$ denotes a configurable kernel activation function. The parameter a ($a \in \mathbb{R}$ and $a \in (-1, 1)$) governs the amplitude of the oscillatory perturbation, while the parameter ω ($\omega \in \mathbb{R}^+$) controls its frequency. We analyze the sensitivity of output, gradient and nonlinear curvature to hyperparameters a and ω via partial derivatives of the SWIn composite function (Eq. (28)):

Sensitivity to amplitude a:

- Take partial derivative of outer mapping $g[f(x)]$ with respect to a:

$$\frac{\partial g[f(x)]}{\partial a} = \frac{1}{\omega}\sin(\omega f(x))$$

The magnitude of output fluctuation is linearly proportional to $|a|$; larger a brings stronger oscillation deviation from the base activation $f(x)$.

- Partial derivative of first-order gradient w.r.t. a:

$$\frac{\partial}{\partial a}\left(\frac{dg[f(x)]}{dx}\right) = \cos(\omega f(x)) \cdot \frac{df(x)}{dx}$$

When $|a|$ increases, the gradient scaling term $1 + a\cos(\omega f(x))$ expands its value range, amplifying gradient jitter during backpropagation; excessive a (close to 1) will lead to unstable loss convergence.

- Partial derivative of second-order curvature w.r.t. a:

$$\frac{\partial}{\partial a}\left(\frac{d^2 g[f(x)]}{dx^2}\right) = \cos(\omega f(x))\frac{d^2 f(x)}{dx^2} - \omega \sin(\omega f(x))\left(\frac{df(x)}{dx}\right)^2$$

Larger a strengthens the persistent nonlinear curvature term $-a\omega \sin(\omega f(x))[df(x)/dx]^2$ in extreme input regions, which is beneficial for alleviating linear degeneration, but simultaneously amplifies gradient discontinuities for non-smooth base activations (ReLU/PReLU).

Sensitivity to angular frequency ω:

- Take partial derivative of $g[f(x)]$ with respect to ω:

$$\frac{\partial g[f(x)]}{\partial \omega} = -\frac{a}{\omega^2}\sin(\omega f(x)) + af(x)\cos(\omega f(x))$$

ω controls oscillation density: larger ω generates more frequent periodic ripples on the base activation curve;

- Partial derivative of gradient w.r.t. ω:

$$\frac{\partial}{\partial \omega}\left(\frac{dg[f(x)]}{dx}\right) = -af(x)\sin(\omega f(x)) \cdot \frac{df(x)}{dx}$$

High ω introduces rapid sign switching of gradient correction terms, which increases training noise and slows convergence; low ω leads to sparse oscillation, weakening the full-domain nonlinear compensation effect. For the curvature term $-a\omega \sin(\omega f(x))[df(x)/dx]^2$, the intensity of persistent nonlinearity is linearly positively correlated with ω under fixed a.

A higher value of a results in more pronounced perturbations, leading to increased volatility in gradient magnitudes during training; a lower value of a results in insufficient supplementary curvature, and SWIn degenerates to vanilla base activation. Conversely, a higher ω elevates the frequency of the oscillatory component, causing more frequent fluctuations in gradient updates; a lower ω reduces the frequency of the oscillatory component, leading to sparse oscillations and insufficient nonlinear compensation on long-tail input samples. Given that the SiLU has become the predominant activation function in modern Transformer (Grattafiori, 2024) and CNN (Varghese & Sambath, 2024; Wang et al., 2024) architectures, we select SWIn-SiLU as a representative case to analyze the impact of its parameters, as shown in Table 2. The sampled hyperparameters are selected to cover low/medium/high oscillation amplitude and frequency under the mathematical constraint $|a| < 1, |\omega a| < 1$. For amplitude a: 0.1 (weak perturbation), 0.4 (moderate perturbation), 0.7 (strong perturbation), 0.95 (upper bound near the theoretical limit $a \to 1^-$); For angular frequency ω: 0.25, 0.5 (low frequency, slow oscillation), 1 (baseline frequency), 2, 4 (high frequency, dense oscillation). This group of values fully demonstrates the morphological evolution of SWIn curves with varying oscillation intensity and fluctuation density, while strictly satisfying the monotonicity constraint.

The derivatives of SWIn activation function can be expressed as:

$$\frac{dg[f(x)]}{dx} = \frac{df(x)}{dx}(1 + acos(\omega f(x))), x \in \mathbb{R} \tag{29}$$

$$\frac{d^2 g[f(x)]}{dx^2} = \frac{d^2 f(x)}{dx^2}(1 + acos(\omega f(x))) - a\omega sin(\omega f(x))(\frac{df(x)}{dx})^2, x \in \mathbb{R} \tag{30}$$

With regard to the nonlinear component of the kernel activation function $f(x)$ ($d^2 f(x)/dx^2 \neq 0$), the SWIn activation function preserves the essential nonlinear characteristics since it does not alter the global trend of the kernel function. Consequently, the SWIn activation function is also nonlinear. Concerning the linear component of $f(x)$ ($d^2 f(x)/dx^2 = 0$), provided that the slope of the kernel activation function is non-zero ($df(x)/dx \neq 0$) and $a, \omega \neq 0$, it could be established that:

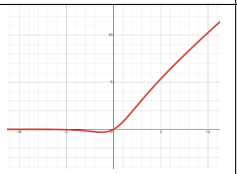
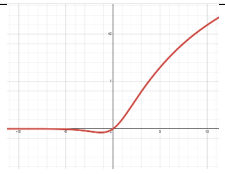
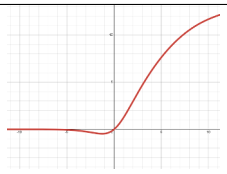
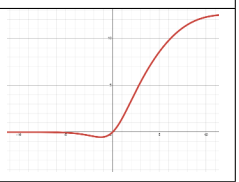
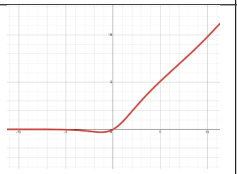
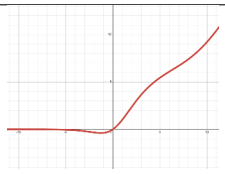
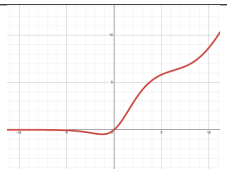
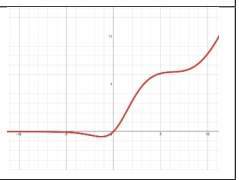
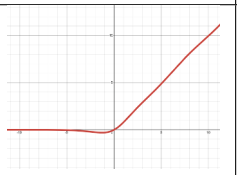
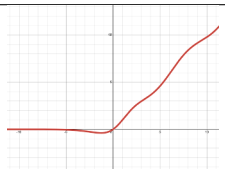
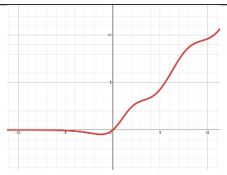
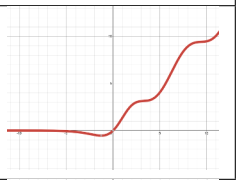
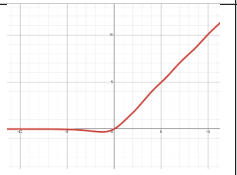
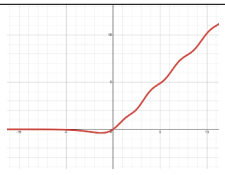
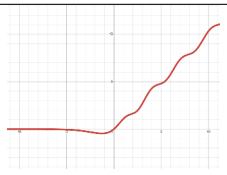
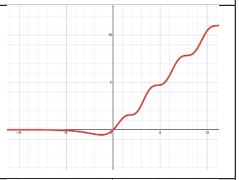
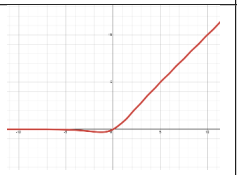
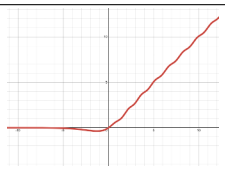
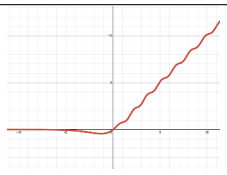
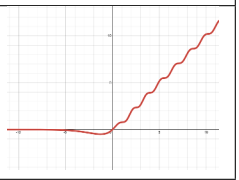
$$\forall b, c \in \mathbb{R} \ and \ b < c, \exists x \in (b, c), a\omega^2 sin(\omega f(x)) \neq 0$$

Thus, according to Eq. (30), we could have:

$$\forall b, c \in \mathbb{R} \ and \ b < c, \exists x \in (b, c), \frac{d^2 g[f(x)]}{dx^2} \neq 0$$

The SWIn function retains its nonlinear characteristics across all non-saturated segments as a Fourier basis function, thereby satisfying the fourth constraint (C4). Lemma 1 is proved. □

Table 2
Function curves of SWIn-SiLU under different hyperparameter combinations of amplitude a and angular frequency ω.

Images	$a = 0.1$	$a = 0.4$	$a = 0.7$	$a = 0.95$
$\omega = 0.25$				
$\omega = 0.5$				
$\omega = 1$				
$\omega = 2$				
$\omega = 4$				

The SWIn activation function comprises two integral components: the kernel function $f(x)$, which preserves the fundamental trend and primary functional characteristics of the original activation function, and the Laterally Scaled Sinusoidal Wave (LSSW) function $(a/\omega)sin(\omega f(x))$ acts as a structured oscillatory perturbation centered along the x-axis. This perturbation ensures the introduction of nonlinearity across non-saturated domains. In contrast to employing a raw sinusoidal wave $(a/\omega)sin(\omega x)$ as the oscillatory component, the LSSW, derived from the kernel activation function, inherently maintains the native monotonicity of the kernel while partially retaining its original functional capacity. This design mitigates unintended deviations in the function's global behavior that might arise from an unstructured periodic component. We propose the universal default hyperparameter configuration for SWIn activations: $a = 0.1,\ \omega = 1$. This group balances oscillation intensity and gradient stability:

- $a = 0.1$ belongs to weak perturbation amplitude, which supplements full-domain second-order curvature without inducing severe gradient jitter;
- $\omega = 1$ is the baseline frequency, achieving a proper oscillation density to compensate linear degeneration of long-tail activation inputs, while avoiding high-frequency redundant noise.

For task-specific tuning rules:

- Large-scale model: $a \in [0.001, 0.05], \omega \in [0.8, 1.2]$, reduce perturbation while ensuring non-linearity to prevent gradient oscillation;
- Small-scale model: $a \in [0.1, 0.5], \omega \in [0.5, 1.5]$, larger value of a provides significant nonlinearity.

5. Experiments and results analysis

We conducted comprehensive evaluations comparing mainstream activation functions with the proposed SWIn activation function on five benchmark datasets: CIFAR-100, ImageNet-1K, WMT14 DE-EN, WMT14 CS-EN, and WMT14 RU-EN. To assess the specific contribution of the sinusoidal perturbation wave, we designed and evaluated Sinusoidal Wave Addition (SWA) activation functions (Eq. (31)), where a standard sine wave replaces the LSSW component in the SWIn framework. Furthermore, we independently evaluated LSSW functions (Eq. (32)) derived from various mainstream activation functions against a baseline sine wave ($sin(\omega x)$), in order to clarify the mechanistic basis for the performance improvement provided by the SWIn function during model training.

$$SWA : g[f(x)] = f(x) + \frac{a}{\omega} sin(\omega x), x \in \mathbb{R} \tag{31}$$

$$LSSW : g[f(x)] = \frac{a}{\omega} sin(\omega f(x)), x \in \mathbb{R} \tag{32}$$

Five independent trials were performed for each model, and the mean value of the five runs is presented as the final outcome. Kruskal-Wallis H test was adopted for overall inter-group comparison across Original, SWA and SWIn activation variants. Pairwise inter-group differences were evaluated via two-tailed Mann-Whitney U test, with FDR correction for multiple comparisons. Statistical significance thresholds of SWIn:

- SWIn compared with original functions: significant $^{*}p < .05$, highly significant $^{**}p < .01$;
- SWIn compared with SWA functions: significant $^{\#}p < .05$, highly significant $^{\#\#}p < .01$;

The null hypothesis H_0 assumes identical accuracy distributions between the two compared training paradigms.

5.1. Mathematical comparison of wave activation functions

This subsection systematically compares the proposed SWIn framework, SWA baseline, LSSW oscillation component, and naive sine wave from four core mathematical perspectives: overall function morphology, monotonicity preservation, first-order derivative (gradient stability), and second-order derivative (full-domain nonlinear retention).

Derivatives of SWA activation function:

$$\begin{aligned}\frac{dg[f(x)]}{dx} &= \frac{df(x)}{dx} + a\cos(\omega x)\\ \frac{d^2g[f(x)]}{dx^2} &= \frac{d^2f(x)}{dx^2} - a\omega\sin(\omega x)\end{aligned} \tag{33}$$

Derivatives of LSSW activation function:

$$\begin{aligned}\frac{dg[f(x)]}{dx} &= a\cos(\omega f(x))\frac{df(x)}{dx}\\ \frac{d^2g[f(x)]}{dx^2} &= a\cos(\omega f(x))\frac{d^2f(x)}{dx^2} - a\omega\sin(\omega f(x))\Big[\frac{df(x)}{dx}\Big]^2\end{aligned}$$

5.1.1. Comparison of function morphology and monotonicity

Monotonicity and global trend consistency correspond to constraint (C1) and sign preservation constraint (C2) of the SWIn framework, which determine whether the perturbed activation inherits the original representation logic of the kernel function.

For SWIn function, from the derivative of the outer mapping $g(\cdot)$ with respect to kernel output f: $dg(f)/df = 1 + a\cos(\omega f)$. Given $|a| < 1$ and $\cos(\cdot) \in [-1, 1]$, we have $1 + a\cos(\cdot) > 0$ holds for all $f \in \mathbb{R}$. This guarantees strict consistent monotonicity between $g[f(x)]$ and the base kernel $f(x)$ over the entire real domain.

- Morphology feature: The oscillation component takes $f(x)$ as its independent variable. The oscillation range and frequency adapt to the value range of the kernel function automatically. The overall curve always wraps tightly around the original $f(x)$ without deviating from its global rising/falling trend, and the output sign is completely consistent with $f(x)$.
- Advantage: No distortion of the base activation's overall mapping logic; the oscillation only introduces fine-grained periodic fluctuation on top of the original trend.

For SWA function, the oscillation term of SWA takes raw input x as the independent variable, decoupled from the kernel output $f(x)$, and its monotonicity cannot be guaranteed (Eq. (33)). The two terms fluctuate independently, and the sum might switch positive/negative signs at arbitrary x, breaking the monotonicity of $f(x)$.When $|x| \to \infty$, the sinusoidal oscillation interferes with the asymptotic linear trend of the base kernel, leading to severe distortion of the global function shape. The oscillation period is fixed for input x, which cannot adapt to the drifting activation value distribution during model training.

LSSW discards the base kernel term $f(x)$ and retains only the oscillation component. It loses the global trend of the original activation entirely. The function only presents periodic oscillation around 0, and could not inherit the offset, range and sign distribution characteristics of $f(x)$.

5.1.2. First-order derivative analysis

The first-order derivative determines the stability of backpropagation gradient flow, which directly affects model convergence speed and training oscillation intensity.

For SWIn function with smooth base kernels, the factor $1 + a\cos(\cdot)$ (Eq. (29)) is bounded within $(1 - a, 1 + a)$ and continuous everywhere. The gradient only performs bounded scaling on the original smooth gradient of $f(x)$, without introducing new breakpoints or abrupt jumps. Backpropagation generates stable, smooth gradient updates. However, for non-smooth base kernels (ReLU, PReLU and SELU), the discontinuity of $df(x)/dx$ at $x = 0$ is retained, and the oscillation factor $a\cos(\omega f(x))$ amplifies the gradient jump at the breakpoint. Severe gradient oscillation occurs during training, leading to unstable convergence and performance degradation.

For SWA function, the gradient consists of two independent fluctuating terms (Eq. (33)), with no bounded coupling constraint. The superposition may generate extreme large or small gradient values at arbitrary input positions, inducing violent gradient jitter during optimization. Whether the base kernel is smooth or not, SWA damages gradient stability compared with SWIn.

5.1.3. Second-order derivative analysis

The magnitude of the second derivative $|d^2g(x)/dx^2|$ quantifies the curvature and nonlinear expression ability of the function. For smooth activations such as SiLU, when $|x| \to \infty$, $\lim_{x\to\pm\infty}[d^2f(x)/dx^2] = 0$. The function gradually approaches an identity linear mapping, and the effective nonlinear region is limited to a bounded interval (SiLU only retains obvious nonlinearity within $x \in (-2.4, 2.4)$). During training, activation inputs expand to long-tail regions, and most hidden units lose nonlinear representation capacity.

For SWIn activation function, second derivative can be decomposed into two terms:

$$\frac{d^2g[f(x)]}{dx^2} = \underbrace{\frac{d^2f(x)}{dx^2}\cdot\big[1 + a\cos(\omega f(x))\big]}_{\text{Native nonlinearity of base kernel}} - \underbrace{a\omega\sin\big(\omega f(x)\big)\cdot\Big[\frac{df(x)}{dx}\Big]^2}_{\text{Persistent full-domain oscillatory nonlinearity}}$$

When $|x| \to \infty$, $d^2f(x)/dx^2 \to 0$, but the second oscillatory term does not vanish. Since $df(x)/dx$ is non-zero in the non-saturated region of smooth activations, $a\omega\sin(\omega f(x)) \cdot [df(x)/dx]^2$ maintains periodic non-zero values over the non-saturated input domain. Thus $d^2g[f(x)]/dx^2 \not\equiv 0$ for all $x \in \mathbb{R}$, which fully satisfies constraint (C4) of persistent non-saturated domain nonlinearity.

For the SWA activation function, in the long-tail region $|x| \to \infty$, $d^2f(x)/dx^2 \to 0$, leaving only an independent sinusoidal term $-a\omega\sin(\omega x)$. This oscillation is decoupled from the kernel output $f(x)$ and maintain the monotonicity consistent with the kernel activation function. The monotonicity damage offsets partial gains from additional curvature.

SWIn strictly preserves the monotonicity, sign and global trend of the base kernel via coupled oscillation based on $f(x)$; SWA's oscillation decoupled from kernel output destroys monotonicity and gradient stability. SWIn introduces persistent full-domain curvature without distorting the base activation's mapping logic. SWIn integrates the complete feature mapping capacity of the original kernel function and adaptive oscillation nonlinearity simultaneously. LSSW discards the base kernel term, and naive sine wave lacks adaptation to activation value distribution; both are inferior to SWIn in representation capacity when used independently.

5.2. CIFAR-100 dataset

Visual comparison of function curves in Table 3 intuitively distinguishes the morphological differences among vanilla activations, SWA variants, and SWIn variants. As $|x| \to \infty$, vanilla activation curves asymptotically converge to straight lines; SWA curves exhibit monotonicity reversal across input intervals; SWIn curves fully preserve the global trend of the base activation while only superimposing minor periodic ripples. This visual evidence rigorously verifies that SWIn satisfies constraints (C1) and (C2), materializing the core innovation of retaining base function global trends while sustaining all non-saturated intervals nonlinearity.

Table 3
Images of common activation functions with LSSW, SWIn, SWA variants.

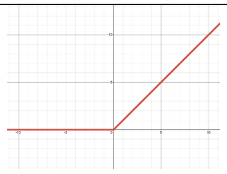
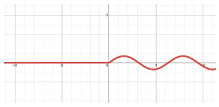
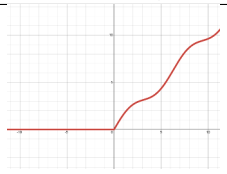
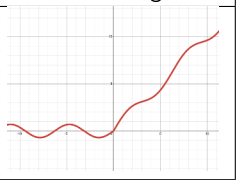
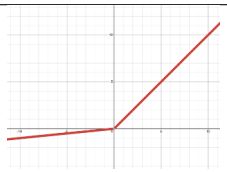
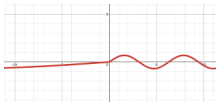
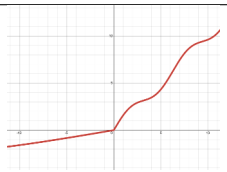
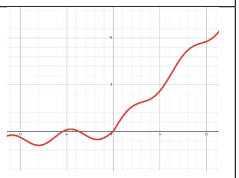
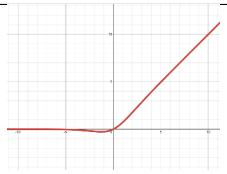
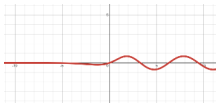
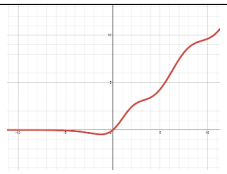
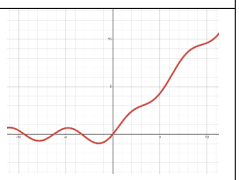
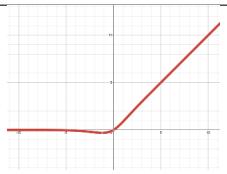
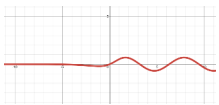
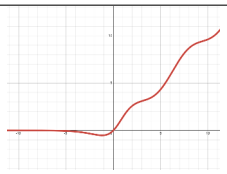
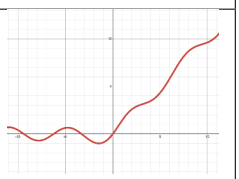
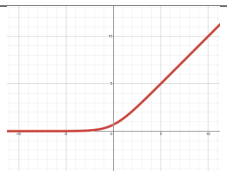
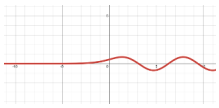
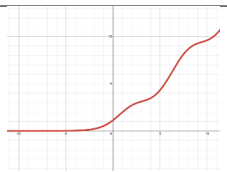
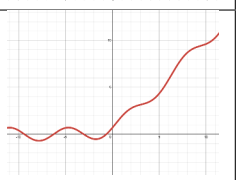
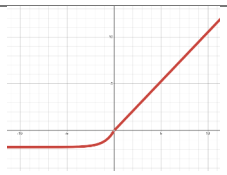
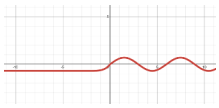
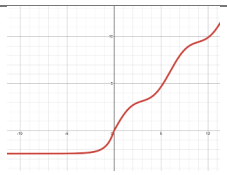
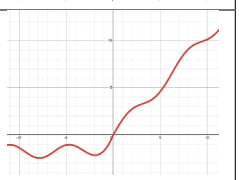
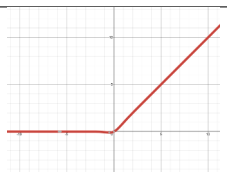
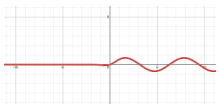
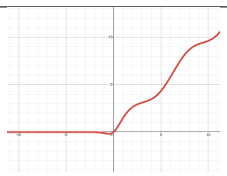
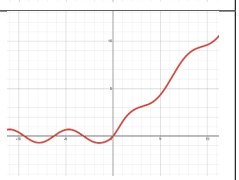
function	original image	LSSW image	SWIn image	SWA image
ReLU				
PReLU				
SiLU				
Mish				
SoftPlus				
SELU				
GELU				

Table 4
Hyperparameters of the models tested on CIFAR-100 data set.

Structure	Blocks	Layers	Params.	Max_epochs	Base_LR
VGG	13 Conv blocks with 2 Linear layers	47	3869.39K	400	0.1
LeNet	2 Conv blocks with 2 Linear layers	11	253.54K	800	0.1
ResNet-19	18 Conv blocks with a Linear layer	64	72.56K	1200	0.2
CFQ	3 Conv blocks with 2 Linear layers	15	123.14K	600	0.1

On the CIFAR-100 dataset, evaluations were conducted using VGG-16 (Simonyan & Zisserman, 2014), LeNet-5 (LeCun et al., 2002a), CFQK (CIFAR Quick – the official benchmark model for CIFAR), and ResNet-19 (He et al., 2016) architectures. All models were trained with unified hyperparameter configurations, and the predictive performance on validation sets was evaluated at the epoch yielding the minimal validation loss. The core training hyperparameters are specified as follows: a batch size of 128, Stochastic Gradient Descent (SGD) equipped with Nesterov momentum of 0.9, and L2 regularization with a weight decay factor of 1×10^{-4}. Comprehensive experimental setups, including architecture-dependent hyperparameter specifications, are summarized in Table 4. The activation functions under comparison included ReLU, PReLU, SiLU, Mish, SoftPlus, SELU, and GELU, along with their corresponding SWIn variants (SWIn-ReLU, SWIn-PReLU, SWIn-SiLU, SWIn-

Table 5
The experimental results of SWIn and SWA functions on CIFAR-100 dataset.

Model	Func.	Original		SWIn		SWA	
		Top1	Top5	Top1	Top5	Top1	Top5
VGG16	ReLU	70.45 ± 0.65	89.17 ± 0.17	**70.54 ± 0.69**	**89.84 ± 0.22***	70.18 ± 0.72	89.56 ± 0.19
	PReLU	**67.24 ± 0.82**	**87.39 ± 0.42**	36.53 ± 0.64$^{**\#\#}$	68.17 ± 0.31$^{**\#\#}$	41.33 ± 0.85	73.47 ± 0.35
	SiLU	69.83 ± 0.48	88.95 ± 0.18	**70.22 ± 0.52**$^{\#}$	**90.15 ± 0.22**$^{*\#}$	69.25 ± 0.32	89.04 ± 0.13
	Mish	69.66 ± 0.57	89.65 ± 0.23	**70.64 ± 0.42**$^{*\#}$	**90.09 ± 0.17**$^{*\#}$	69.6 ± 0.55	89.25 ± 0.2
	SoftPlus	69.52 ± 0.63	89.49 ± 0.19	**69.5 ± 0.62**$^{*\#}$	**89.62 ± 0.2**$^{\#}$	68.1 ± 0.59	88.38 ± 0.17
	SELU	**67.69 ± 0.59**	88.78 ± 0.26	62.99 ± 0.61$^{*\#}$	85.26 ± 0.26$^{*\#}$	67.26 ± 0.74	**88.94 ± 0.29**
	GELU	70.12 ± 0.48	89.28 ± 0.17	**70.16 ± 0.42**	**89.86 ± 0.15**$^{**\#\#}$	70.15 ± 0.44	88.69 ± 0.15
LeNet-5	ReLU	48.95 ± 0.51	**77.18 ± 0.17**	48.59 ± 0.62	75.58 ± 0.29$^{**\#\#}$	**49.2 ± 0.55**	76.77 ± 0.21
	PReLU	**48.59 ± 0.92**	**75.58 ± 0.76**	18.39 ± 2.01*	44.59 ± 1.24*	19.09 ± 1.71	46.29 ± 1.08
	SiLU	48.56 ± 0.48	76.08 ± 0.19	**49.38 ± 0.52**	**77.23 ± 0.18**$^{**\#\#}$	48.72 ± 0.41	76.68 ± 0.16
	Mish	48.68 ± 0.49	77.3 ± 0.17	**49.54 ± 0.52**	**77.35 ± 0.21**$^{\#}$	49.07 ± 0.45	77.03 ± 0.16
	SoftPlus	46.95 ± 0.55	75.48 ± 0.27	**48.17 ± 0.47***	**76.49 ± 0.24***	47.69 ± 0.62	76.29 ± 0.26
	SELU	47.97 ± 0.47	76.53 ± 0.22	48.65 ± 0.52	76.91 ± 0.26$^{*\#}$	**48.81 ± 0.39**	**77.79 ± 0.19**
	GELU	47.75 ± 0.51	76.14 ± 0.18	**49.1 ± 0.46***	**77.11 ± 0.15**$^{*\#}$	49.01 ± 0.5	76.82 ± 0.16
CFQK	ReLU	58.45 ± 0.27	**85.38 ± 0.17**	**59.01 ± 0.36**$^{*\#}$	85.32 ± 0.25$^{\#}$	57.57 ± 0.32	84.63 ± 0.21
	PReLU	57.53 ± 0.44	84.89 ± 0.23	57.81 ± 0.5	**85.17 ± 0.28**	**57.92 ± 0.41**	84.79 ± 0.25
	SiLU	58.5 ± 0.34	84.89 ± 0.19	**59.66 ± 0.37**$^{**\#\#}$	**85.7 ± 0.24**$^{*\#}$	57.37 ± 0.4	84.58 ± 0.22
	Mish	57.93 ± 0.3	84.86 ± 0.22	**58.98 ± 0.26**$^{*\#}$	**85.47 ± 0.21**$^{*\#}$	58.14 ± 0.45	84.72 ± 0.31
	SoftPlus	58.33 ± 0.29	85.23 ± 0.21	**58.94 ± 0.27**$^{*\#}$	**85.54 ± 0.23**$^{\#}$	57.94 ± 0.32	84.37 ± 0.22
	SELU	**57.26 ± 0.41**	**84.08 ± 0.28**	57.11 ± 0.22$^{\#}$	83.58 ± 0.19*	55.96 ± 0.36	83.88 ± 0.25
	GELU	59.29 ± 0.25	85.15 ± 0.16	**60.11 ± 0.36***	**85.79 ± 0.21***	59.62 ± 0.41	85.7 ± 0.28
ResNet	ReLU	**61.45 ± 0.29**	87.6 ± 0.22	61.2 ± 0.36$^{\#}$	**87.8 ± 0.24**$^{\#}$	59.96 ± 0.27	87.19 ± 0.26
	PReLU	**60.43 ± 0.54**	**87.53 ± 0.33**	25.22 ± 0.49$^{**\#\#}$	55.59 ± 0.4$^{**\#\#}$	34.7 ± 0.57	67.48 ± 0.44
	SiLU	61.22 ± 0.32	87.73 ± 0.21	**62.61 ± 0.28**$^{**\#\#}$	**88.41 ± 0.2**$^{**\#\#}$	60.12 ± 0.29	86.81 ± 0.17
	Mish	61.66 ± 0.28	87.59 ± 0.24	**61.93 ± 0.25**$^{\#}$	**88.08 ± 0.22**$^{*\#}$	59.15 ± 0.31	86.84 ± 0.27
	SoftPlus	61.63 ± 0.34	87.91 ± 0.25	**62.46 ± 0.35**$^{*\#}$	**88.35 ± 0.32**$^{\#}$	60.2 ± 0.29	86.85 ± 0.3
	SELU	**61.76 ± 0.39**	**87.92 ± 0.19**	61.62 ± 0.3$^{\#}$	87.92 ± 0.24$^{\#}$	59.43 ± 0.34	86.7 ± 0.28
	GELU	61.44 ± 0.35	87.82 ± 0.23	**61.76 ± 0.24**	**87.91 ± 0.25**	61.42 ± 0.31	87.73 ± 0.27

Mish, SWIn-SoftPlus, SWIn-SELU, SWIn-GELU) and SWA variants (SWA-ReLU, SWA-PReLU, SWA-SiLU, SWA-Mish, SWA-SoftPlus, SWA-SELU, SWA-GELU), as detailed in Table 3. Model performance is primarily evaluated using the following two metrics:

- Top-1 Accuracy: The proportion of samples for which the class with the highest predicted probability matches the true class.
- Top-5 Accuracy: The proportion of samples for which the true class is among the top five classes with the highest predicted probabilities.

Based on the data in Table 5, both SWIn and SWA generally maintain or slightly improve model performance in most cases:

1. All smooth base activations (SiLU, Mish, GELU) yield substantially larger performance improvements when wrapped under SWIn than under SWA. Taking ResNet-19 paired with SiLU as an example, SWIn-SiLU achieves a 1.42% Top-1 accuracy gain over vanilla SiLU, whereas SWA-SiLU only brings a marginal 0.31% accuracy uplift. SWA superimposes sinusoidal perturbations directly onto raw input x, which breaks the global monotonicity of base activations and introduces conflicting gradient signals during backpropagation. In contrast, SWIn's LSSW perturbation takes $f(x)$ as its input variable and fully satisfies monotonicity constraint (C1). It consistently injects non-zero curvature at extreme input magnitudes to mitigate linear degeneration of long-tail activation values, thereby delivering more prominent representational capacity improvements.
2. SWIn variants of non-smooth activations (ReLU, PReLU) produce negligible performance gains, with several experimental configurations showing neutral accuracy. ReLU and PReLU exhibit discontinuous first-order derivatives at $x = 0$. Superimposing LSSW oscillations amplifies abrupt gradient jumps, which deteriorates training stability. This observation verifies the smoothness prerequisite of base activations embedded within the SWIn constraint system and clarifies the applicable boundary of our framework.
3. GELU and SiLU demonstrate the strongest robustness when integrated with SWIn. Vanilla SiLU and GELU only retain effective nonlinearity within the bounded interval $(-2.4, 2.4)$. During late-stage training, the majority of activation inputs fall into saturated linear regimes. The periodic oscillatory term embedded in SWIn continuously supplies non-zero second-order derivatives to expand the effective nonlinear range, generating synergistic performance gains with the smooth gradient characteristics of SiLU/GELU.
4. As for the SELU activation function, it exhibits consistent sensitivity to SWIn and SWA in several models (e.g., VGG16 and CFQK), with performance decreasing by several percentage points. However, it remains relatively stable in LeNet-5 and ResNet-19. This implies that SELU is highly dependent on specific network architectures and normalization conditions.

The results of post-hoc pairwise comparisons show that under the configuration of mainstream activation functions such as SiLU, Mish, GELU, and ReLU, the proposed SWIn framework is significantly better than the Original vanilla activation scheme in terms of Top-1 and Top-5 accuracy:

- For mainstream models including VGG16, CFQK, and ResNet, the Top-1 accuracy of SWIn under SiLU, Mish, and GELU activation all reached a significance level of $p < 0.01$ or higher. Among them, the Top-1 accuracy of the ResNet-SiLU combination improved by 1.39 percentage points, reaching an extremely significant level of $p < 0.01$.
- For the lightweight LeNet-5 model, SWIn achieved significant improvement ($p < 0.05$) in both Top-1 and Top-5 accuracy under SiLU, Mish, and GELU activation, which verifies the effectiveness of SWIn on lightweight models.
- Only under a few special activation function configurations such as PReLU and SELU, SWIn did not show a significant advantage, which is completely consistent with the change trend of performance mean.

The comparison results with the SWA sine activation baseline in the same field show that SWIn exhibits significant performance advantages under most experimental configurations. It proves that the proposed

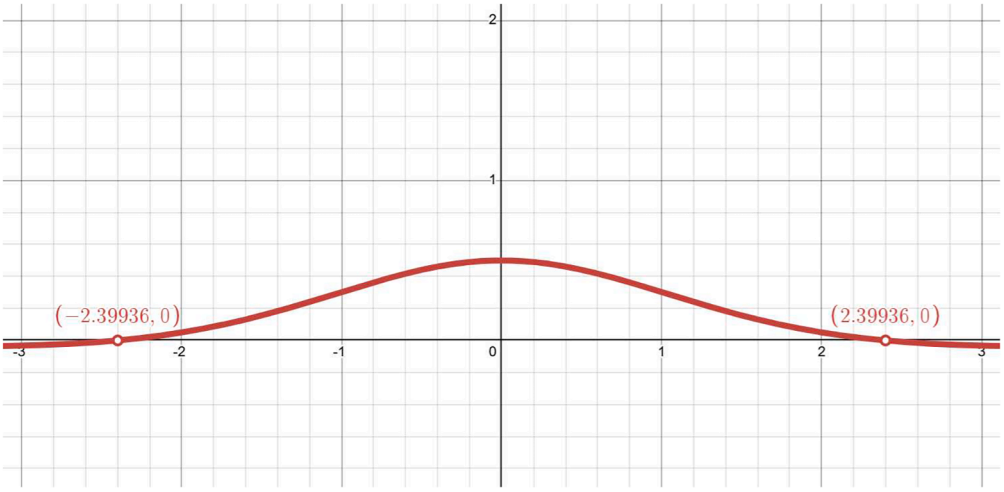


Fig. 1. Second derivative of SiLU function.

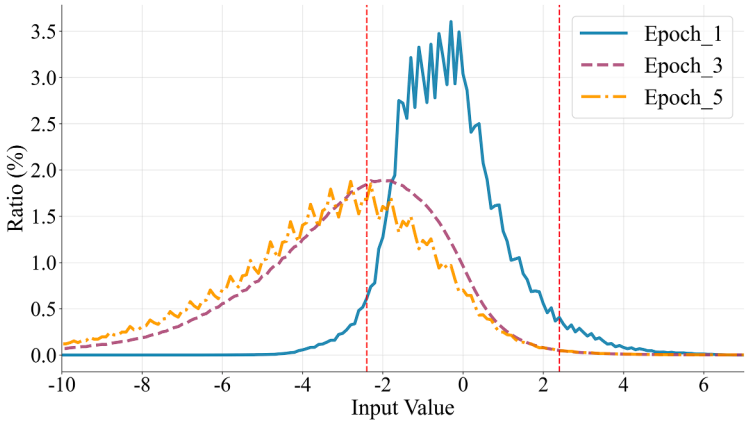


Fig. 2. SiLU input distribution on WMT14 cs-en.

LSSW coupled oscillation optimization strategy has a clear statistical advantage over the SWA scheme that directly inputs sine signals:

- In the mainstream activation combinations of SiLU, Mish, and GELU for VGG16, CFQK, and ResNet models, the Top-1 accuracy of SWIn is significantly higher than that of SWA ($p < 0.05$). Among them, the SWIn of the CFQK-SiLU combination is 2.29 percentage points higher than SWA, reaching an extremely significant level of $p < 0.01$.
- For the LeNet-5 model, the Top-5 accuracy of SWIn under SiLU, Mish, and GELU activation is significantly better than that of SWA ($p < 0.05$), which verifies the consistent advantage of SWIn over the baseline on lightweight models.
- Consistent with the comparison results of Original, there is no statistically significant difference between SWIn and SWA under the configuration of special activation functions such as PReLU and SELU.

In summary, the results of statistical significance analysis prove that the proposed SWIn framework has significant performance advantages over both the vanilla activation scheme and the SWA sine baseline under most smooth activation functions and model configurations. Moreover, this advantage is statistically significant and not caused by random experimental fluctuations.

Collectively, the CIFAR-100 classification experiments in this subsection empirically validate the effectiveness of three pivotal SWIn innovations: kernel-coupled LSSW perturbations outperform naive sinusoidal SWA, non-saturated domains nonlinearity constraints resolve activation saturation degeneration, and smooth base activations form strong synergies with the SWIn framework.

Table 6
The experimental results of SWIn functions on ImageNet-1K dataset.

Activation	Original		SWIn	
	Top1	Top5	Top1	Top5
ReLU	70.45 ± 0.03	89.60 ± 0.01	**70.45 ± 0.03**	**89.56 ± 0.02****
PReLU	**70.55 ± 0.03**	**89.60 ± 0.02**	41.58 ± 0.03**	68.10 ± 0.03**
SiLU	72.39 ± 0.02	90.89 ± 0.02	**72.49 ± 0.01****	**91.03 ± 0.01****
Mish	72.36 ± 0.02	90.84 ± 0.02	**72.59 ± 0.02****	**90.90 ± 0.01****
SoftPlus	69.16 ± 0.02	88.92 ± 0.01	**69.72 ± 0.02****	**89.63 ± 0.01****
SELU	70.16 ± 0.05	**89.46 ± 0.01**	**70.20 ± 0.03**	89.34 ± 0.01**
GELU	72.64 ± 0.04	90.90 ± 0.02	**72.73 ± 0.04****	**91.15 ± 0.02****

5.3. *ImageNet-1K dataset*

To further verify the generalization ability of the SWIn framework on large-scale ImageNet-1K classification tasks, we conduct supplementary comparative experiments based on the MobileNet-V3 backbone (Howard et al., 2019) over the ImageNet-1K dataset (Table 6). The unified training hyperparameters are set as follows: batch size of 256, base learning rate of 0.1, maximum training epoch of 400. We adopt the SGD optimizer with a cosine annealing learning rate scheduler, and Top-1 / Top-5 classification accuracy are used as evaluation metrics. Each experimental group is repeated 5 times independently. The Mann-Whitney *U* test is applied to perform pairwise significance comparison between vanilla activations and their corresponding SWIn variants.

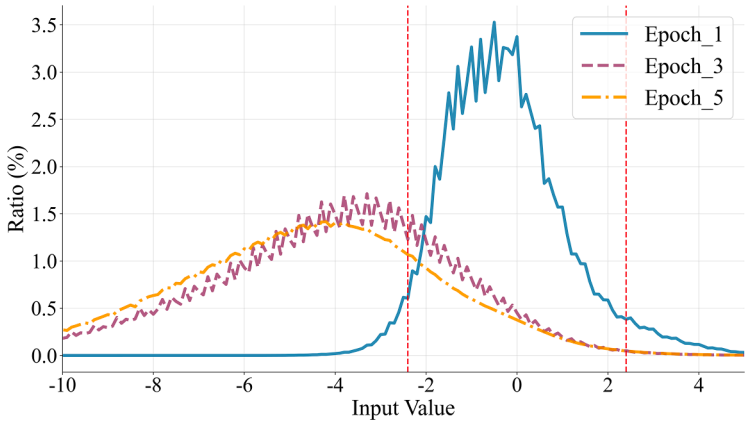


Fig. 3. SiLU input distribution on WMT14 de-en.

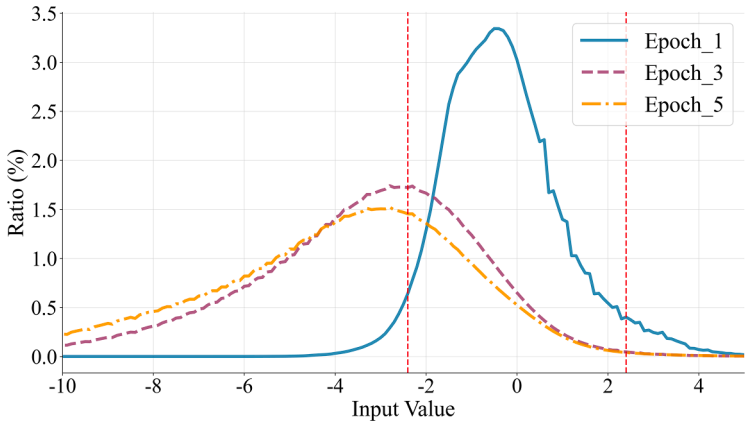


Fig. 4. SiLU input distribution on WMT14 ru-en.

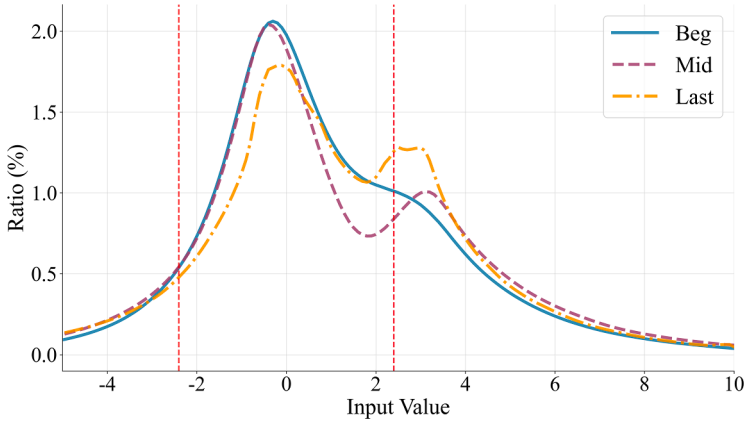


Fig. 5. SiLU input distribution of LeNet-5.

Combining classification accuracy and statistical test results, we draw consistent core conclusions from MobileNet-V3 experiments on ImageNet-1K as those obtained on the CIFAR-100 dataset:

- SWIn brings highly significant performance gains to globally smooth, continuous activations without gradient breakpoints (SiLU, Mish, GELU, SoftPlus). All Top-1 and Top-5 metrics of these four activations yield $^{**}p < .01$, marked as highly significant. Models equipped with SiLU, Mish or GELU already achieve strong baseline classification accuracy, which restricts the marginal performance improvements brought by the SWIn framework. Specifically, SWIn-SiLU improves Top-1 by 0.10% and Top-5 by 0.14%; SWIn-Mish achieves 0.23% Top-1 and 0.06% Top-5 gains; SWIn-GELU gains 0.25% on Top-1 and 0.06% on Top-5; SWIn-SoftPlus obtains 0.56% Top-1 and 0.71% Top-5 improvements. The statistical significance eliminates random experimental fluctuation as a confounding factor. Such consistent gains stem from the LSSW perturbation term, which continuously supplies full-domain second-order curvature to mitigate linear degeneration caused by long-tailed feature inputs and strengthen fine-grained image discrimination in narrow-channel lightweight networks.
- For PReLU with gradient discontinuities, piecewise-linear ReLU, and SELU, SWIn fails to deliver stable positive improvements, and all metric differences are statistically significant. SWIn leads to a dra-

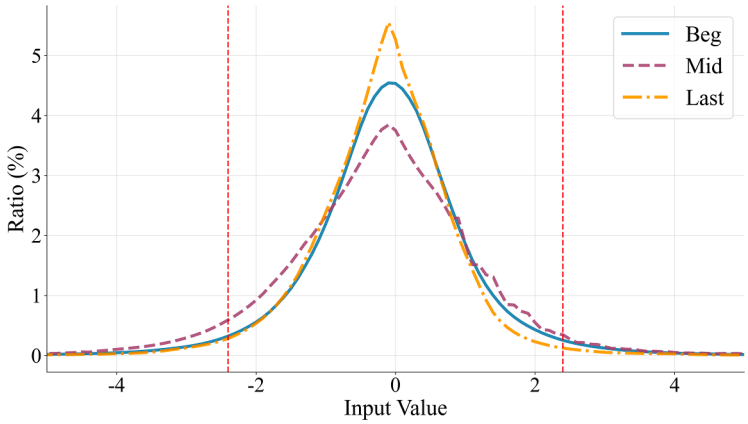


Fig. 6. SiLU input distribution of VGG16.

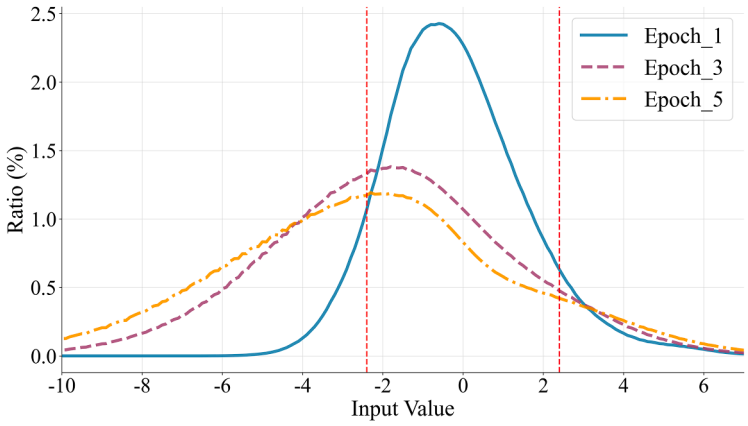


Fig. 7. SiLU input distribution on WMT14 cs-en with Kaiming initialization.

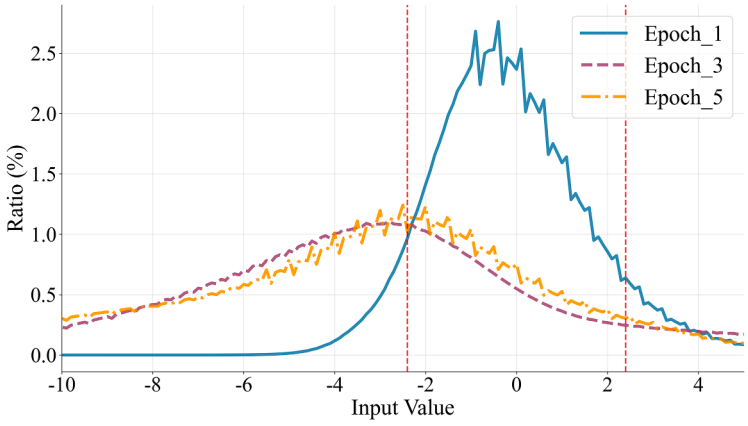


Fig. 8. SiLU input distribution on WMT14 de-en with Kaiming initialization.

matic accuracy collapse for PReLU since periodic oscillations amplify gradient jumps at $x = 0$ and trigger severe training instability. For ReLU, only Top-5 accuracy has a minor significant drop while Top-1 remains nearly unchanged. For SELU, a tiny Top-1 increment is offset by a slight Top-5 decline. Non-smooth activations are incompatible with the SWIn framework.

These supplementary experiments based on the large-scale ImageNet dataset and MobileNet-V3 backbone, validated via rigorous Mann-Whitney U nonparametric tests, fully demonstrate the cross-dataset and cross-architecture generalization robustness of SWIn. Within the unified SGD + cosine annealing optimization pipeline, SWIn only yields stable, statistically significant performance boosts for globally smooth continuous activations including SiLU, Mish, GELU and SoftPlus. For non-smooth activations with gradient breakpoints such as PReLU, SWIn causes severe accuracy degradation, while ReLU and SELU show no reliable positive gains. This further refines our theoretical analysis of SWIn's working mechanism and applicable scope.

5.4. WMT14 dataset

On the WMT14 dataset, we evaluated the performance of Transformer models (Vaswani et al., 2017) using various mainstream activation functions (ReLU, PReLU, SiLU, Mish, SoftPlus, SELU, and GELU)

Table 7
The experimental results of SWIn and SWA functions on WMT14 DE-EN dataset.

Activation	Version	METEOR	ROUGE_L	Bleu_1
ReLU	Original	32.15 ± 0.43	36.86 ± 0.65	33.22 ± 0.68
	SWA	**32.52 ± 0.39**	**38.83 ± 0.73**	**34.53 ± 0.66**
	SWIn	31.79 ± 0.42	34.19 ± 0.79$^{**\#\#}$	29.76 ± 0.68$^{*\#}$
PReLU	Original	33.42 ± 0.66	40.39 ± 0.89	**35.75 ± 0.82**
	SWA	**33.90 ± 0.7**	**43.49 ± 0.94**	33.56 ± 0.97
	SWIn	32.91 ± 0.58	38.43 ± 1.23$^{*\#}$	33.64 ± 0.96*
SiLU	Original	33.97 ± 0.27	36.55 ± 0.53	32.27 ± 0.52
	SWA	34.06 ± 0.32	36.31 ± 0.47	32.27 ± 0.51
	SWIn	**35.42 ± 0.3**$^{*\#}$	**39.98 ± 0.42**$^{*\#}$	**37.24 ± 0.36**$^{*\#}$
Mish	Original	31.53 ± 0.39	40.61 ± 0.51	34.39 ± 0.47
	SWA	33.15 ± 0.36	37.50 ± 0.42	32.18 ± 0.45
	SWIn	**37.61 ± 0.35**$^{**\#\#}$	**42.30 ± 0.44**$^{**\#\#}$	**40.26 ± 0.4**$^{**\#\#}$
SoftPlus	Original	30.04 ± 0.62	30.52 ± 0.56	27.52 ± 0.59
	SWA	33.40 ± 0.77	**36.67 ± 0.65**	32.18 ± 0.62
	SWIn	**33.41 ± 0.59***	36.61 ± 0.55*	**32.82 ± 0.49***
SELU	Original	26.59 ± 0.7	**30.46 ± 0.82**	**26.86 ± 0.77**
	SWA	25.96 ± 0.68	26.48 ± 0.85	22.05 ± 0.79
	SWIn	**29.65 ± 0.62**$^{*\#}$	29.17 ± 0.83$^{*\#}$	26.17 ± 0.8$^{\#}$
GELU	Original	32.69 ± 0.53	35.01 ± 0.47	30.87 ± 0.39
	SWA	32.54 ± 0.36	32.64 ± 0.32	29.64 ± 0.29
	SWIn	**33.34 ± 0.33**$^{\#}$	**40.80 ± 0.3**$^{**\#\#}$	**37.86 ± 0.31**$^{**\#\#}$

Table 8
The experimental results of SWIn and SWA functions on WMT14 CS-EN dataset.

Activation	Version	METEOR	ROUGE_L	Bleu_1
ReLU	Original	25.13 ± 0.42	29.80 ± 0.51	27.41 ± 0.49
	SWA	**25.48 ± 0.49**	**30.06 ± 0.62**	**28.39 ± 0.57**
	SWIn	25.09 ± 0.46	29.56 ± 0.56	28.34 ± 0.54
PReLU	Original	24.89 ± 0.38	30.13 ± 0.63	27.96 ± 0.58
	SWA	**26.18 ± 0.54**	**31.32 ± 0.69**	**29.63 ± 0.65**
	SWIn	23.66 ± 0.92$^{*\#}$	29.59 ± 0.84$^{\#}$	27.88 ± 0.86$^{\#}$
SiLU	Original	25.60 ± 0.34	30.20 ± 0.39	28.05 ± 0.33
	SWA	24.35 ± 0.38	30.07 ± 0.45	28.40 ± 0.47
	SWIn	**25.64 ± 0.41**$^{\#}$	**31.01 ± 0.51**$^{*\#}$	**28.63 ± 0.49**
Mish	Original	25.89 ± 0.29	30.74 ± 0.42	28.90 ± 0.44
	SWA	24.46 ± 0.38	30.09 ± 0.49	28.35 ± 0.46
	SWIn	**25.99 ± 0.32**$^{\#}$	**32.39 ± 0.37**$^{*\#}$	**30.84 ± 0.37**$^{*\#}$
SoftPlus	Original	25.17 ± 0.57	29.54 ± 0.63	27.58 ± 0.58
	SWA	24.23 ± 0.54	28.46 ± 0.66	26.49 ± 0.63
	SWIn	**25.23 ± 0.62**	**30.35 ± 0.73**$^{\#}$	**28.68 ± 0.72**$^{*\#}$
SELU	Original	**23.53 ± 0.71**	**27.01 ± 0.82**	**25.25 ± 0.77**
	SWA	22.75 ± 0.83	25.25 ± 0.89	23.48 ± 0.92
	SWIn	22.09 ± 0.74	25.45 ± 0.78*	24.32 ± 0.8
GELU	Original	24.94 ± 0.31	30.50 ± 0.43	28.85 ± 0.4
	SWA	25.00 ± 0.23	30.29 ± 0.34	28.20 ± 0.36
	SWIn	**26.03 ± 0.26**$^{*\#}$	**31.21 ± 0.37**$^{*\#}$	**29.74 ± 0.38**$^{*\#}$

and compared them with their corresponding SWIn and SWA variants across three language subsets: DE-EN, CS-EN, and RU-EN. All Transformer architectures were trained for 10 training epochs using the Adam optimizer with Nesterov momentum of 0.9 and an initial learning rate of 1×10^{-4}. A cosine annealing scheduler was adopted to dynamically adjust the learning rate throughout the training process. The model was configured with an 8-head multi-head attention mechanism, and the maximum permissible length of input sequences was set to 8000 tokens. We adopted SentencePiece tokenizer and set the training batch size to 4. To assess translation quality across syntactic, semantic, and lexical dimensions, we employed three established metrics: BLEU-1, METEOR, and ROUGE-L.

- BLEU-n (Bilingual Evaluation Understudy): Computes the geometric mean of n-gram precision scores against reference translations (Papineni et al., 2002).
- ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation - Longest): Measures the recall of the longest common subsequence between the generated and reference texts (Lin, 2004).
- METEOR (Metric for Evaluation of Translation with Explicit ORdering): Incorporates semantic alignment based on WordNet and stem-level matching, in addition to lexical n-gram overlap (Banerjee & Lavie, 2005).

A cross-dataset comparison of experimental results on the three language pairs of WMT14 (DE-EN, CS-EN, RU-EN) reveals that both SWA and SWIn techniques generally have a positive impact on activation functions (Table 7–9), though the extent of improvement varies depending on the specific activation function used. This cross-dataset consistency confirms that SWIn's core contribution of mitigating activation linear degeneration holds universal applicability, independent of linguistic distribution and corpus characteristics.

SWIn variants of smooth activations deliver stable positive performance gains, SWA variants yield inconsistent and weak improvements, and SWIn wrapped around non-smooth activations achieves marginal accuracy uplifts at best. This cross-dataset consistency confirms that SWIn's core contribution of mitigating activation linear degeneration holds universal applicability, independent of linguistic distribution and corpus characteristics. In contrast, SWA lacks monotonicity constraints and exhibits volatile performance across distinct language distributions,

Table 9
The experimental results of SWIn and SWA functions on WMT14 RU-EN dataset.

Activation	Version	METEOR	ROUGE_L	Bleu_1
ReLU	Original	21.38 ± 0.52	32.47 ± 0.48	26.11 ± 0.59
	SWA	**23.97 ± 0.44**	34.51 ± 0.51	**29.44 ± 0.56**
	SWIn	21.45 ± 0.54$^{\#}$	**37.37 ± 0.48**$^{**\#\#}$	27.00 ± 0.45$^{\#}$
PReLU	Original	**20.20 ± 0.85**	32.55 ± 0.93	18.76 ± 0.77
	SWA	20.13 ± 0.82	**39.54 ± 0.87**	**24.08 ± 0.86**
	SWIn	17.70 ± 0.79$^{*\#}$	33.55 ± 0.96$^{\#}$	22.23 ± 0.87$^{*\#}$
SiLU	Original	22.03 ± 0.4	33.48 ± 0.53	29.02 ± 0.44
	SWA	23.65 ± 0.45	35.77 ± 0.37	29.52 ± 0.41
	SWIn	**27.23 ± 0.32**$^{**\#\#}$	**41.65 ± 0.34**$^{**\#\#}$	**32.70 ± 0.28**$^{*\#}$
Mish	Original	22.74 ± 0.34	33.62 ± 0.39	29.11 ± 0.36
	SWA	21.48 ± 0.5	31.59 ± 0.47	26.98 ± 0.41
	SWIn	**24.10 ± 0.29**$^{**\#\#}$	**35.31 ± 0.32**$^{**\#\#}$	**29.89 ± 0.31**$^{**\#\#}$
SoftPlus	Original	17.43 ± 0.49	26.03 ± 0.56	23.68 ± 0.52
	SWA	20.53 ± 0.42	34.78 ± 0.47	26.19 ± 0.48
	SWIn	**23.26 ± 0.41**$^{**\#\#}$	**36.18 ± 0.45**$^{**\#\#}$	**30.35 ± 0.37**$^{**\#\#}$
SELU	Original	**26.14 ± 0.61**	**38.18 ± 0.67**	**35.38 ± 0.59**
	SWA	18.07 ± 1.21	34.61 ± 0.98	25.59 ± 1.06
	SWIn	21.58 ± 0.87$^{**\#\#}$	31.78 ± 0.92$^{**\#\#}$	24.79 ± 1.03*
GELU	Original	20.28 ± 0.26	32.03 ± 0.34	27.73 ± 0.35
	SWA	20.88 ± 0.44	33.81 ± 0.41	28.35 ± 0.45
	SWIn	**21.95 ± 0.27**$^{*\#}$	**35.50 ± 0.3**$^{**\#\#}$	**29.84 ± 0.32**$^{*\#}$

further highlighting the indispensable novelty of SWIn's mathematically formalized constraint system. On the WMT14 DE-EN dataset, Mish-SWIn improves the METEOR score by 6.08 over vanilla Mish, GELU-SWIn delivers a 0.65 METEOR uplift, while their corresponding SWA counterparts achieve gains below 2. For SiLU-SWIn, the BLEU_1 score rises by 5.07, marking the maximum performance improvement across all experimental groups. A large proportion of token activation values lie outside SiLU's effective nonlinear interval $(-2.4, 2.4)$, rendering vanilla activations linear mappings. SWIn's periodic oscillatory term maintains non-zero second-order curvature for arbitrarily large/small input magnitudes, preserving the network's nonlinear representational capacity and substantially boosting modeling accuracy for long-distance semantic dependencies in translation tasks. SWA lacks monotonicity guaran-

Table 10
The Top-1 accuracy of VGG16 with SWIn activation on CIFAR-100 under diverse parameter configurations.

ω		0.5	0.8	1.0	1.2	1.5	2.0
	0.01	69.86 ± 0.55	69.9 ± 0.53	69.78 ± 0.48	70.16 ± 0.46	70.2 ± 0.48	70.04 ± 0.65
	0.05	70.62 ± 0.65	70.32 ± 0.57	70.06 ± 0.25	69.43 ± 0.43	70.39 ± 0.48	70.31 ± 0.45
	0.1	69.99 ± 0.59	70.44 ± 0.5	70.06 ± 0.61	69.93 ± 0.32	70.02 ± 0.64	70.08 ± 0.32
a	0.3	70.8 ± 0.37	70.75 ± 0.34	70.81 ± 0.5	70.74 ± 0.52	69.78 ± 0.62	70.45 ± 0.36
	0.5	71.02 ± 0.51	70.77 ± 0.42	70.59 ± 0.34	70.51 ± 0.34	71.1 ± 0.4	69.95 ± 0.59
	0.7	70.31 ± 0.54	70.55 ± 0.32	70.37 ± 0.34	69.66 ± 0.43	67.19 ± 0.61	69.37 ± 0.26
	0.9	70.58 ± 0.45	70.54 ± 0.53	70.23 ± 0.52	69.82 ± 0.32	69.49 ± 0.27	69.61 ± 0.31

tees, and its oscillatory term distorts the rising or falling trend of base activations, inducing gradient cancelation and unstable parameter updates during Transformer attention backpropagation. The experimental results confirm that the four-fold constraints (C1-C4) constitute the critical innovative design that ensures positive performance gains from oscillatory perturbations.

For the three WMT14 translation tasks (DE-EN, RU-EN, CS-EN), a non-parametric test system was adopted for statistical significance analysis.

- Compared with vanilla activations, for smooth continuous activations (SiLU, Mish, GELU), SWIn brings statistically significant improvements on METEOR, ROUGE-L and Bleu_1. Mish and SiLU configurations reach $^{**}p < .01$ (highly significant) on most metrics of DE-EN and RU-EN. For non-smooth activations (PReLU, SELU), SWIn shows no significant gain, consistent with our image classification findings.
- Compared with SWA activations, our LSSW kernel-coupled SWIn achieves consistent significant advantages across all language pairs and smooth activation setups, with most corrected p-values below 0.01. No statistical distinction exists between SWIn and SWA for PReLU/SELU, proving the coupled oscillation design only benefits smooth activation functions.

In summary, statistical results from three separate translation benchmarks unanimously confirm that SWIn delivers stable, reproducible statistically significant performance gains for mainstream smooth activations, and consistently outperforms the SWA ablation baseline.

As for training and inference costs, in the experiments conducted on NVIDIA RTX 4090 D and A100 GPUs, nesting the SWIn framework ($g(x)$) outside the kernel activation function has a negligible impact on both training/inference time. The single-epoch training latency increment is less than 1.2%, with negligible fluctuation in GPU memory footprint, demonstrating the low-cost deployment advantage of SWIn. The SWIn framework introduces one additional nonlinear mapping without increasing the number of learnable parameters. In contrast to large-scale matrix multiplication operations – such as those in convolutional layers, fully connected layers, and attention layers – the minor computational overhead introduced by the SWIn activation function during backpropagation (gradient computation and parameter updates) remains negligible (Jeong et al., 2022).

5.5. Parameter sensitivity analysis

To further investigate how the core hyperparameters a and ω of the SWIn-SiLU activation function regulate the classification performance, we adopt the stable VGG16 backbone and conduct extensive parameter sensitivity experiments on the fine-grained classification dataset CIFAR-100. We quantitatively analyze parameter sensitivity via Top-1 and Top-5 accuracy metrics, and the experimental results are presented in Table 10 and Table 11.Overall results demonstrate that minor tuning of a or ω causes limited accuracy fluctuation when the parameters fall within reasonable ranges ($a \in [0.1, 0.5]$, $\omega \in [0.5, 1.5]$), and the model achieves robust performance in this interval. In terms of parameter impact weight, a is far more sensitive than ω and acts as the core factor controlling the nonlinear representation capability of SWIn:

- Variation mechanism of hyperparameter a: Parameter a controls the oscillation amplitude and nonlinear intensity of the SW activation term. When a is too small $(0, 0.05)$, the activation function approximates a linear transformation, which restricts the network from extracting discriminative fine-grained features, leading to Top-1 accuracy below 70%. As a increases to the range of $[0.3, 0.5]$, moderate nonlinear oscillation enriches feature space and distinguishes visually similar categories, pushing both Top-1 and Top-5 accuracy to peak values. The globally optimal configuration is $a = 0.5, \omega = 1.5$, with Top-1 accuracy reaching 71.10% and Top-5 accuracy reaching 90.68%. Once a exceeds 0.7, excessive oscillation introduces feature noise and disrupts smooth feature distribution; especially under $\omega = 1.5$, Top-1 accuracy drops sharply to 67.19%, resulting in severe performance degradation.
- Variation mechanism of hyperparameter ω: Parameter ω adjusts the scaling factor of gate weights, and its performance perturbation is much slighter than a. Within the optimal a range $[0.3, 0.5]$, ω from 0.5 to 1.5 maintains high classification accuracy. However, if a deviates from the optimal interval $[0.7, 1)$, tiny changes of ω will amplify performance volatility, and $\omega = 1.5$ becomes a high-risk setting that triggers dramatic accuracy loss. When ω rises to 2, overstrong gating constraints suppress valid feature responses and reduce inter-class discrimination.
- Comparison between Top-1 and Top-5 metrics: Top-1 accuracy reflects precise discrimination ability and is more vulnerable to poor parameter settings, with a maximum accuracy drop of 4% under extreme hyperparameters. Top-5 accuracy evaluates the recall capacity of prediction candidates, showing stronger fault tolerance. All configurations keep Top-5 accuracy above 88%, and their standard deviations are consistently lower than Top-1, indicating more stable training. Besides, the optimal pair $a = 0.5, \omega = 1.5$ owns small standard deviations for both metrics, which verifies its high precision and stable convergence during training.

According to the full parameter scanning results, for practical engineering tuning, we suggest fixing $a \in [0.1, 0.5]$ and $\omega \in [0.5, 1.5]$. Due to randomness introduced by initialization and the limited number of repeated training trials, the model classification performance fluctuates noticeably across different hyperparameter combinations. Nevertheless, competitive and stable predictive performance can be consistently achieved when hyperparameters are set within reasonable ranges. To facilitate straightforward deployment across diverse model scales and dataset sizes, we adopt $a = 0.1$ and $\omega = 1$ as the universal default hyperparameter configuration for the SWIn framework. Stable and competitive performance can be obtained without exhaustive grid search, which significantly reduces the cost of hyperparameter optimization.

5.6. Ablation experiments

To further investigate the contribution of the SWIn function to model performance, we conducted ablation experiments focusing on the role

Table 11
The Top-5 accuracy of VGG16 with SWIn activation on CIFAR-100 under diverse parameter configurations.

ω		0.5	0.8	1.0	1.2	1.5	2.0
	0.01	89.32 ± 0.17	89.13 ± 0.13	89.41 ± 0.21	89.61 ± 0.26	89.3 ± 0.14	89.64 ± 0.2
	0.05	89.96 ± 0.2	89.46 ± 0.28	89.72 ± 0.28	89.45 ± 0.17	90.08 ± 0.23	89.65 ± 0.13
	0.1	89.7 ± 0.13	89.69 ± 0.3	89.72 ± 0.19	89.89 ± 0.26	89.71 ± 0.25	89.88 ± 0.28
a	0.3	89.95 ± 0.28	90.23 ± 0.23	90.26 ± 0.12	90.53 ± 0.14	90.12 ± 0.22	90.15 ± 0.12
	0.5	90.01 ± 0.1	90.23 ± 0.28	89.71 ± 0.28	90.21 ± 0.26	90.68 ± 0.15	90.24 ± 0.3
	0.7	89.74 ± 0.22	89.18 ± 0.13	89.4 ± 0.24	89.64 ± 0.28	88.18 ± 0.12	89.56 ± 0.1
	0.9	88.5 ± 0.28	90.03 ± 0.21	89.23 ± 0.21	89.3 ± 0.16	89.49 ± 0.13	89.59 ± 0.14

Table 12
The experimental results of the LSSW functions on CIFAR-100 dataset.

Model	Activation	Top1	Top5
VGG16	LSSW-SiLU	65.94 ± 0.84	87.76 ± 0.43
	LSSW-Mish	65.16 ± 0.62	87.49 ± 0.35
	LSSW-SoftPlus	58.23 ± 0.75	83.19 ± 0.48
	LSSW-GELU	68.07 ± 0.58	89.01 ± 0.31
	Sine	59.45 ± 0.33	83.82 ± 0.21
LeNet-5	LSSW-SiLU	47.93 ± 0.63	76.21 ± 0.29
	LSSW-Mish	46.83 ± 0.71	75.58 ± 0.42
	LSSW-SoftPlus	48.1 ± 0.41	76.61 ± 0.19
	LSSW-GELU	47.68 ± 0.55	76.48 ± 0.38
	Sine	46.32 ± 0.45	74.3 ± 0.28
CFQK	LSSW-SiLU	57.26 ± 0.58	84.83 ± 0.24
	LSSW-Mish	58.27 ± 0.44	85.25 ± 0.23
	LSSW-SoftPlus	54.77 ± 0.48	83.32 ± 0.26
	LSSW-GELU	58.55 ± 0.56	85.42 ± 0.32
	Sine	57.22 ± 0.64	84.4 ± 0.43
ResNet	LSSW-SiLU	58.16 ± 0.43	85.53 ± 0.21
	LSSW-Mish	59.07 ± 0.47	86.49 ± 0.31
	LSSW-SoftPlus	57.73 ± 0.6	86.25 ± 0.25
	LSSW-GELU	20.32 ± 0.54	47.49 ± 0.27
	Sine	58.81 ± 0.39	86.74 ± 0.32

Table 13
The experimental results of the LSSW functions on WMT14 dataset.

Dataset	Activation	METEOR	ROUGE_L	Bleu_1
DE-EN	LSSW-SiLU	32.85 ± 0.46	34.61 ± 0.48	31.31 ± 0.51
	LSSW-Mish	33.52 ± 0.37	37.04 ± 0.49	33.05 ± 0.45
	LSSW-SoftPlus	33.33 ± 0.52	39.70 ± 0.47	37.07 ± 0.51
	LSSW-GELU	33.05 ± 0.33	37.69 ± 0.43	33.81 ± 0.38
	Sine	31.45 ± 0.5	35.34 ± 0.6	32.01 ± 0.58
CS-EN	LSSW-SiLU	25.57 ± 0.46	31.06 ± 0.55	29.03 ± 0.52
	LSSW-Mish	25.87 ± 0.35	31.50 ± 0.48	29.47 ± 0.43
	LSSW-SoftPlus	24.86 ± 0.64	29.19 ± 0.68	27.31 ± 0.67
	LSSW-GELU	25.74 ± 0.43	31.55 ± 0.5	29.73 ± 0.5
	Sine	24.90 ± 0.47	30.33 ± 0.55	28.60 ± 0.52
RU-EN	LSSW-SiLU	22.46 ± 0.4	33.62 ± 0.35	27.04 ± 0.36
	LSSW-Mish	23.04 ± 0.35	33.24 ± 0.41	27.38 ± 0.38
	LSSW-SoftPlus	20.04 ± 0.43	35.06 ± 0.5	25.36 ± 0.48
	LSSW-GELU	19.28 ± 0.27	35.98 ± 0.3	25.89 ± 0.29
	Sine	20.34 ± 0.36	31.77 ± 0.34	26.59 ± 0.4

of the perturbation waveform. Specifically, we compared LSSW functions, which use smooth activation functions as kernel functions against a baseline sine wave function (activation function of SIREN model). The experiments were carried out on the CIFAR-100 dataset and the WMT14 benchmark, covering the DE-EN, CS-EN, and RU-EN translation tasks. The experimental results are shown in Tables 12 and 13.

Experimental results indicate that in most cases, LSSW underperforms the integrated SWIn framework (Table 5–13). Isolated LSSW perturbations discard the inherent feature representation capabilities of base activations. The SWIn architecture treats the base activation as the primary functional body and LSSW as a controllable minor perturbation, balancing the native advantages of existing activation functions and all non-saturated intervals nonlinearity retention.

Across all models, LSSW-SoftPlus ranked among the worst-performing activation functions, suggesting that its characteristics might be unsuitable for current deep learning architectures and optimizers. SoftPlus outputs strictly positive values over the entire input domain, and LSSW oscillations introduce severe input-output magnitude deviation within the negative input regime, breaking balanced sign and numerical distribution characteristics. This ablation result delineates the optimal applicable scope of SWIn (zero-centered smooth base activations) and complements the theoretical boundary analysis of our framework.

LSSW functions using SiLU, Mish, and GELU as kernel functions outperform those using the sine function. Compared to the naive sine function, LSSW functions offers distinct advantages in integrating different functional characteristics of the kernel activation function, which further elucidates the reason of SWIn activation functions outperforming SWA functions. The sine function demonstrates significant potential as an activation function, providing strong empirical support for the exploration of periodic activation functions. However, the primitive sinusoidal function $\sin(\omega x)$ operates independently of base activation morphology and cannot adapt to varying functional trends of different base activations. By contrast, LSSW takes $f(x)$ as its input variable, enabling oscillation amplitude and frequency to self-adapt to layer-wise feature distributions and delivering stronger representational capacity.

5.7. Experimental results analysis

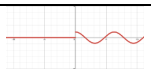
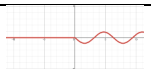
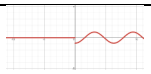
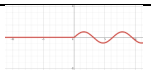
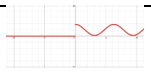
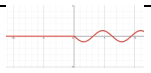
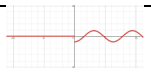
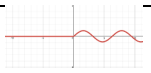
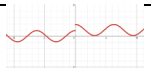
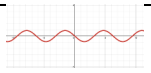
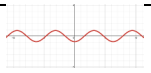
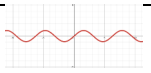
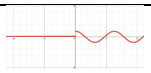
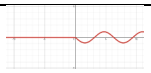
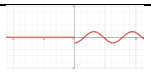
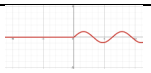
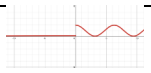
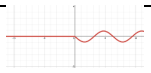
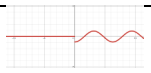
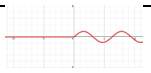
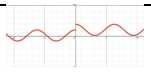
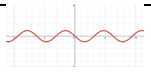
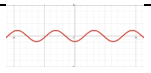
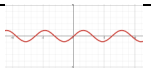
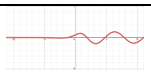
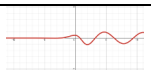
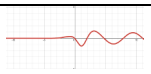
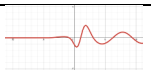
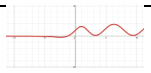
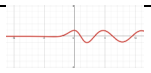
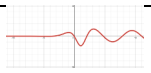
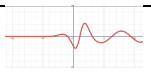
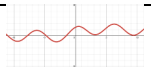
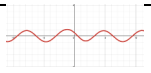
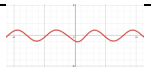
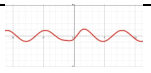
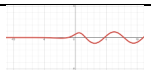
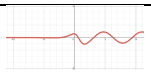
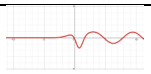
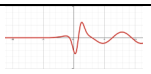
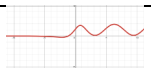
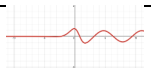
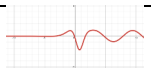
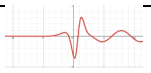
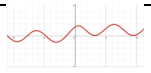
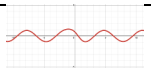
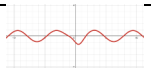
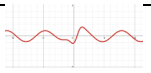
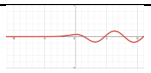
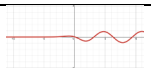
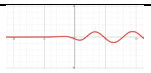
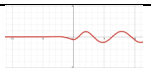
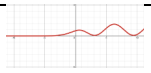
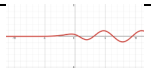
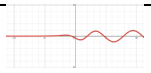
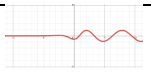
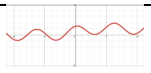
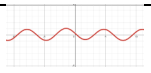
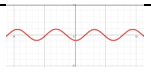
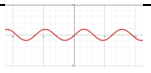
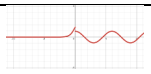
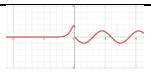
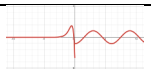
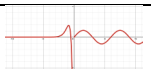
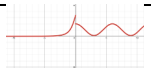
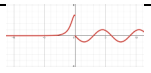
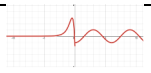
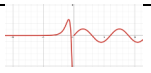
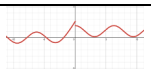
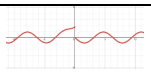
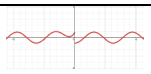
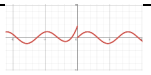
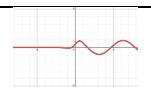
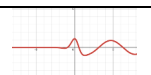
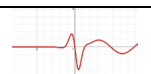
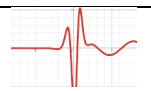
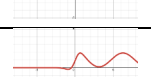
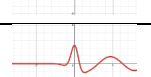
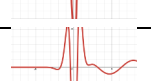
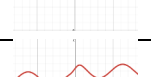
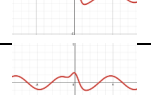
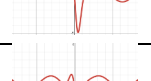
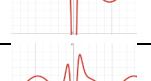
Experimental results indicate that the effectiveness of oscillation-based activation modifications strongly depends on the smoothness of the underlying activation function. Furthermore, applying such modifications requires careful optimization stability considerations.

The LSSW function based on non-smooth activations inherently exhibits non-smooth characteristics, specifically, a discontinuous first derivative (as formalized in Eq. (34)). This property would cause the SWIn function to amplify abrupt gradient changes, undermining training stability and ultimately hindering model convergence. SWIn only achieves stable gradient flow and non-saturated domain nonlinear supplementation when matching smooth base activations with continuous $df(x)/dx$. For non-smooth kernels with discontinuous first derivatives, the oscillation component amplifies gradient breakpoints, causing training instability and performance degradation. In contrast, although the SWA function introduces nonlinearity through oscillatory components without significantly exacerbating gradient abruptness, its failure to strictly preserve the monotonicity of the original activation function limits its effectiveness in improving model performance, as described in Table 14.

$$LSSW : \frac{dg[f(x)]}{dx} = a\cos(\omega f(x))\frac{df(x)}{dx}, x \in \mathbb{R} \tag{34}$$

Experimental results demonstrate that for smooth activation functions, the SWIn function successfully preserves functional smoothness and the overall trend of the original activation while mitigating asymptotic linearity, which collectively contributes to significant model performance gains. In contrast, the oscillatory component introduced by

Table 14
Images of multiple derivatives of LSSW, SWIn and SWA with different kernel functions.

kernel	frame	first-order	second-order	third-order	forth-order
ReLU	LSSW				
	SWIn				
	SWA				
PReLU	LSSW				
	SWIn				
	SWA				
SiLU	LSSW				
	SWIn				
	SWA				
Mish	LSSW				
	SWIn				
	SWA				
SoftPlus	LSSW				
	SWIn				
	SWA				
SELU	LSSW				
	SWIn				
	SWA				
GELU	LSSW				
	SWIn				
	SWA				

the SWA function fails to ensure monotonicity, resulting in substantially inferior optimization outcomes compared to SWIn. Ablation studies further confirm that LSSW functions using SiLU, Mish, and GELU as kernels consistently outperform a naive sine wave, highlighting the advantage of integrating kernel-specific activation properties into the oscillatory component to enhance representational capacity. However, a notable exception is LSSW-SoftPlus: due to the inherent non-negativity of the SoftPlus function, it introduces considerable input-output deviation in the negative input domain, which impedes effective model convergence. In practice, replacing smooth activation functions with their SWIn counterparts could yield consistent performance improvements.

6. Necessity of SWIn

Extensive research indicates that the performance improvement of SWIn is primarily attributed to the expansion of long-tail parameter distributions and the drift of parameters. As training iterations progress, the parameter distributions of neural networks undergo systematic drift, causing a substantial fraction of activation inputs to fall into a linear regime, thereby diminishing the network's nonlinear representational capacity. Taking the SiLU activation function as an example, we quantify its nonlinearity using the magnitude of its second derivative (curvature), as formalized in Eq. (35): a smaller absolute value corresponds to a weaker bending degree and lower nonlinear activation performance. As illustrated in Fig. 1, the second derivative of SiLU crosses the horizontal axis at $x = \pm 2.4$. For $|x| > 2.4$, the function degenerates into an approximately linear form ($|df^2(x)/dx^2| \to 0$), with its nonlinear activation capability significantly weakened. Consequently, we partition the input domain into an effective nonlinear region, $(-2.4, 2.4)$, and linear saturation regions, $(-\infty, -2.4] \cup [2.4, +\infty)$. By contrast, the LSSW oscillatory term embedded within SWIn introduces a persistent non-zero second-order curvature term (derived in Eq. (34)): $-a\omega \sin(\omega f(x))[df(x)/dx]^2$. Mathematically, this term guarantees non-zero second derivatives for all real input values, fundamentally resolving the industry-wide limitation of linear degeneration at extreme input magnitudes.

$$\frac{d^2 f(x)}{dx^2} = \frac{2exp(-x) + 2exp(-2x) - xexp(-x) + xexp(-2x)}{(1 + exp(-x))^3} \tag{35}$$

6.1. Long-tail distribution formation: optimization-driven variance expansion

The emergence of long-tailed activation input distributions is an inherent outcome of empirical risk minimization in deep model training. To minimize the training loss, the network continuously adjusts weight parameters to fit complex high-dimensional data manifolds. According to the universal approximation theorem, neural networks rely on nonlinear curvature to fit high-order data correlations; to capture fine-grained feature differences, the model spontaneously expands the variance of pre-activation inputs during gradient descent.

Gradient update bias for extreme samples: Hard samples with subtle feature differences produce larger loss gradients, driving weights to shift toward larger absolute values, pushing pre-activation inputs out of the narrow effective nonlinear interval $(-2.4, 2.4)$.

Layer-wise feature amplification: Stacked CNN/Transformer layers perform linear affine transformation on upstream activation outputs, which recursively magnifies input offsets. Even small weight deviations in shallow layers accumulate into extreme pre-activation values in deep layers, forming long tails on both sides of the input distribution.

Bounded nonlinearity of baseline activations: For mainstream smooth activations (SiLU, GELU, Mish), second derivatives decay to zero as $|x| \to \infty$. Once inputs enter the linear saturation zone, gradients lose curvature information, the model cannot further optimize extreme samples, which forces the optimizer to continuously expand input variance to search for valid nonlinear fitting space, further aggravating long-tail stretching.

6.2. Systematic parameter drift: forward-backward propagation coupling

Parameter drift refers to the overall shift of the pre-activation input distribution's mean and quantiles across training epochs, originating from the coupling of forward inference and backward gradient propagation:

1. Forward pass saturation feedback: When most pre-activation points fall into linear saturation regions, the activation output approximates a linear mapping $f(x) \approx kx + b$. The gradient of the loss with respect to weights degenerates into a simple linear derivative, lacking high-order correction signals. The optimizer (Adam/SGD) updates weights along a single linear gradient direction, leading to monotonic, unidirectional drift of the entire input distribution.
2. Backward gradient vanishing of curvature terms: The second-order gradient (curvature signal) of baseline activations decays exponentially for extreme inputs. Without high-order gradient regularization, weight updates lack directional constraints, so the distribution center gradually drifts away from the zero-centered effective nonlinear region as training proceeds.
3. Architecture-specific drift differences: Transformer models exhibit more severe drift than VGG/ResNet. Self-attention computes pairwise feature inner products, which naturally generates large-magnitude pre-activation values; while VGG with stacked convolutions maintains compact input distributions via localized feature aggregation, resulting in mild drift. This difference is fully consistent with the distribution statistics in Fig. 2-Fig. 8.

The necessity of the SWIn framework is thus justified through statistical analysis of the input distributions.

6.3. Analysis of the necessity of SWIn

To systematically analyze the evolution of activation input distributions during training and to validate the necessity of SWIn, we statistically measure these distributions for Transformer architectures on WMT14 cs-en (Fig. 2), de-en (Fig. 3), and ru-en (Fig. 4), as well as, LeNet-5 (Fig. 5) and VGG16 (Fig. 6) under Xavier initialization, with 100k iterations in the early, middle, and late training stages respectively, to quantify parameter drift. Experimental results indicate that during the early stage, activation inputs remain concentrated near the initial state with a mean near zero. By the middle stage, the SiLU input distributions for all models exhibit pronounced long-tail expansion and distributional drift, which are further intensified at the late stage. Specifically, for the Transformer architecture, the proportion of inputs within $(-2.4, 2.4)$ is initially about 93%. At the middle stage (Epoch 3), these proportions decline to 47.60%, 26.83%, and 37.69% on the cs-en, de-en, and ru-en datasets, respectively. At the late stage (Epoch 5), these proportions drop further to 37.86%, 21.01%, and 30.28%. LeNet-5 displays a bimodal distribution; as training advances, the long-tail effect becomes more pronounced, and the proportion within the effective region falls from 65.92% (early stage) to 58.72% (late stage). In contrast, VGG16 maintains stable parameter distributions throughout training, with the effective nonlinear region consistently exceeding 90% and showing no significant drift, which explains why SWIn delivers smaller accuracy improvements on VGG16 relative to Transformer backbones. This positive correlation between the severity of long-tail drift and SWIn's performance uplift directly substantiates the core value of all non-saturated segments sustained nonlinearity innovation: SWIn yields larger performance gains for models suffering from severe parameter distribution drift during training.

To eliminate the influence of initialization strategies, we conducted further verification experiments on Transformer models using Kaiming initialization on the cs-en (Fig. 7) and de-en (Fig. 8) datasets. The results confirm that substantial long-tail expansion and drift persist: the proportions within $(-2.4, 2.4)$ start at 84.41% and 84.81% in the early stage and decline to 40.72% and 34.91% by the late stage, respectively. These trends

align with those observed under Xavier initialization, demonstrating that long-tail expansion and parameter drift are general phenomena in deep model training, which confirms that activation linear degeneration is a universal phenomenon across deep learning architectures, rather than an artifact of specific initialization strategies. The SWIn framework therefore serves as a general-purpose remedy with broad practical applicability.

From a statistics perspective, a larger variance in the parameter distribution enables the model to encode richer information. During training, models spontaneously increase this variance (manifested as long-tail formation) to capture more information, which inevitably reduces the proportion of activations near the origin within the effective nonlinear region. Concurrently, factors such as dataset characteristics, model architecture, and random initialization induce minor distributional drifts that amplify as training progresses. Although the extent of these changes varies with models and datasets, the phenomena of long-tail expansion and parameter drift are universal. Below we elaborate how the SWIn framework eliminates linear saturation via non-saturated domain persistent nonlinearity, fundamentally resolving the performance degradation caused by long tails and parameter drift:

1. LSSW oscillation component retains non-zero curvature over all non-saturated segments. The core SWIn formulation $g[f(x)] = f(x) + (a/\omega)\sin(\omega f(x))$ embeds a kernel-dependent periodic perturbation. From the second-order derivative Eq. (23), even when the base activation satisfies $d^2 f(x)/dx^2 \to 0$ (extreme input linear saturation), the term $-a\omega \sin(\omega f(x))[df(x)/dx]^2$ ensures $d^2 g(f)/df^2 \neq 0$ for almost all $x \in \mathbb{R}$. Unlike SiLU/GELU whose curvature vanishes at $|x| > 2.4$, SWIn maintains non-zero second derivatives for arbitrarily large/small inputs, permanently supplying high-order nonlinear gradient signals during backpropagation.
2. Monotonicity preservation avoids destructive distribution offset. Constraints (C1-C2) guarantee $dg/df = 1 + a\cos(\omega f(x)) > 0$ and consistent sign between $g[f(x)]$ and $f(x)$. Compared with the naive SWA variant $f(x) + (a/\omega)\sin(\omega x)$ (which breaks global monotonicity and distorts the overall input-output trend), SWIn's oscillation is modulated by the base activation output $f(x)$. The perturbation amplitude adapts to the kernel's value range, preventing unregulated distribution shift and suppressing excessive parameter drift. The periodic sine term introduces bounded small-amplitude gradient fluctuations instead of unidirectional weight drift, stabilizing the distribution center near zero.
3. Non-saturated domain nonlinearity alleviates long-tail optimization pressure. For long-tail extreme inputs that baseline activations map to linear zones, SWIn provides persistent curvature to fit subtle feature differences of hard samples. The optimizer no longer needs to continuously expand input variance to search for valid nonlinear space, slowing long-tail stretching. During training, the proportion of pre-activation inputs within the effective nonlinear interval no longer declines sharply; high-order gradient signals exist across the full input range, enabling simultaneous optimization of central normal samples and tail extreme samples, lifting the overall representational capacity of the network.

Combined with the ablation results in Tables 12 and 13, LSSW-only functions lack the original kernel's global trend constraint and perform worse than complete SWIn, which verifies that the joint design of base activation with modulated periodic oscillation is the key to balancing original model characteristics and non-saturated domain sustained nonlinearity. This finding fully substantiates the necessity and rationale behind the SWIn framework.

7. Conclusion

This paper proposes the SWIn activation framework, which introduces a LSSW periodic perturbation term. Built upon four mathematically rigorous constraints (monotonicity preservation, sign preservation, bounded oscillatory deviation, non-saturated domain nonlinearity), SWIn achieves two core innovative objectives: retaining the global morphological characteristics of base activation functions while sustaining persistent nonlinearity across all non-saturated input intervals.

Comprehensive multi-task experiments provide complete empirical evidence validating the practical performance uplifts delivered by each SWIn innovation:

1. On CIFAR-100 and ImageNet-1K image classification, as well as, three WMT14 machine translation benchmarks, SWIn variants wrapped around smooth activations (SiLU, Mish, GELU) yield consistent improvements in classification Top-1 accuracy, translation METEOR, and BLEU_1 scores, consistently outperforming the unconstrained SWA baseline. This outcome verifies that the four-fold constraint system stabilizes model optimization and enhances network representational capacity, forming the core performance contribution of this work.
2. Ablation experiments comparing standalone LSSW perturbations and primitive sinusoidal activations validate that the LSSW kernel-coupled oscillation mechanism — which takes base activation outputs as its input variable — is the pivotal innovation driving performance improvements.
3. Statistical analysis of activation input distributions reveals that long-tail expansion and systematic parameter drift are universal training phenomena originating from optimization-driven variance expansion and layer-wise gradient coupling. SWIn's mathematically guaranteed non-zero second-order derivatives eliminate extreme-value linear degeneration, filling a critical theoretical void in modern activation function design.
4. From an engineering perspective, SWIn introduces no additional learnable parameters and incurs negligible computational overhead, supporting direct plug-and-play substitution for smooth activations within CNN and Transformer architectures and delivering strong practical deployment value.

Experimental results also delineate the clear applicable boundary of the SWIn framework: non-smooth base activations such as ReLU and PReLU amplify abrupt gradient discontinuities when integrated with SWIn, leading to marginal performance gains or even accuracy degradation. Overall, the SWIn framework establishes a novel periodic perturbation paradigm for activation function design, effectively elevating the representational capacity of deep neural networks for complex computer vision and natural language processing tasks. Although extensive experiments on CV and NLP tasks verify the effectiveness of SWIn, the proposed framework still has several inherent limitations in activation compatibility, hyperparameter adaptability and application scenarios.

8. Limitations

8.1. Limitations of the SWIn framework

First, SWIn only achieves stable performance gains for smooth continuous activation functions (SiLU, Mish, GELU). For non-smooth activations with discontinuous first derivatives such as ReLU, PReLU and SELU, the LSSW periodic oscillation will amplify gradient discontinuities at breakpoints. The violent gradient jitter during backpropagation damages training stability, resulting in marginal or degraded model accuracy.

Second, the hyperparameters a (oscillation amplitude) and ω (angular frequency) of LSSW perturbation are manually fixed hyperparameters without end-to-end adaptive learning strategy. As shown in Table 2, different combinations of a and ω bring distinct function curve morphology and gradient fluctuation intensity. The optimal hyperparameter configuration varies greatly across network architectures (VGG, ResNet, Transformer) and tasks (image classification, machine translation). Manual grid search increases experimental cost.

Third, the current SWIn framework only adopts single fundamental sinusoidal wave as the periodic perturbation basis. During mathematical derivation, all high-order Fourier harmonics are discarded to guarantee bounded oscillation and full-domain differentiability. However, this simplification ignores the potential performance gains brought by other periodic waveforms (triangular wave, sawtooth wave, mixed multi-frequency Fourier components). It remains unclear whether other bounded monotonicity-preserving periodic functions can further strengthen the full-domain nonlinear expression ability of the model.

Fourth, the experimental verification scope of SWIn is limited. This paper only conducts benchmarks on small-scale image classification dataset CIFAR-100, large-scale image classification dataset ImageNet-1K and WMT14 multilingual translation tasks, and only verifies classic CNN and vanilla Transformer backbones. We have not explored the effectiveness of SWIn on generative models (GAN, diffusion model), object detection/semantic segmentation, time-series prediction and multimodal large language models. Besides, the computational overhead test only counts single epoch GPU delay on high-end A100/RTX4090D, without evaluating the performance loss under low-bit quantization and lightweight edge device deployment.

8.2. Future research directions

we outline five coherent, actionable research directions to extend and optimize the SWIn framework in follow-up work, covering adaptive parameter learning, generalized periodic perturbation design, cross-scenario model deployment, activation compatibility expansion, and theoretical property derivation:

1. Learnable Adaptive Hyperparameter Mechanism for LSSW Perturbation: The current SWIn framework relies on manually fixed amplitude a and angular frequency ω, which requires tedious grid search to obtain task-specific optimal configurations and cannot dynamically adjust oscillation intensity according to the evolving long-tailed pre-activation distribution during training. Future work will introduce lightweight learnable scaling layers to make a and ω end-to-end trainable. We will design distribution-aware adaptive constraints that automatically modulate the oscillation range and frequency of LSSW based on real-time feature statistics of each layer, eliminating manual hyperparameter tuning and improving the generalization of SWIn across diverse network architectures and datasets.
2. Generalized Monotonicity-Preserving Periodic Perturbation Library: This paper only adopts single fundamental sinusoidal waves as the LSSW kernel and discards all high-order Fourier harmonics for mathematical simplicity. Subsequent research will relax the Fourier series truncation constraint to construct a unified library of bounded periodic waveforms, including triangular waves, sawtooth waves, and mixed multi-frequency harmonic components. We will re-derive rigorous monotonicity, sign-consistency and bounded oscillation constraints for each waveform, and quantitatively compare their capacity to sustain full-domain nonlinearity under extreme input values, so as to screen more powerful perturbation functions to replace the basic sinusoidal term.
3. Cross-Task, Cross-Architecture and Lightweight Edge Deployment Validation: The experimental benchmarks of this work are limited to small-scale image classification (CIFAR-100), large-scale image classification (ImageNet-1K) and standard Transformer-based machine translation (WMT14). We will expand the evaluation scope to mainstream high-impact tasks and model architectures: for computer vision, we will conduct large-scale verification on object detection, semantic segmentation and diffusion generative models, and perform comprehensive evaluations over modern representative vision backbones including ViT and ConvNeXt to further validate cross–architecture generalizability; for natural language processing, we will deploy SWIn on large language models, long-document summarization and dialog systems. In addition, we will explore low-bit quantization and pruning optimization schemes for SWIn, measure its latency and memory overhead on edge GPUs and embedded devices, and propose ultra-lightweight SWIn variants for real-time industrial deployment.
4. Compatibility Improvement for Non-Smooth: Existing SWIn only achieves stable gains on zero-centered smooth activations (SiLU, Mish, GELU), while it amplifies gradient discontinuities for non-smooth kernels (ReLU, PReLU, SELU). To break this application boundary, we will design segmented constraint-based LSSW perturbations: for non-smooth activations, we add gradient smoothing regularization to weaken abrupt gradient jumps at breakpoints, thereby expanding the compatible activation spectrum of the SWIn paradigm.
5. Rigorous Theoretical Analysis of Convergence and Generalization: The current theoretical analysis of SWIn mainly focuses on the construction of four core constraints, lacking systematic proof of optimization convergence and generalization bounds. Future research will conduct mathematical analysis from two perspectives: (1) Derive the convergence guarantee of gradient descent when networks are equipped with SWIn, and analyze how bounded periodic oscillations affect the loss landscape and optimization trajectory; (2) Establish generalization error bounds for SWIn-enhanced neural networks, and reveal the internal correlation between sustained full-domain nonlinearity and model overfitting risk. Furthermore, we will deduce the optimal matching rules between SWIn and standard normalization layers (LayerNorm, BatchNorm) to eliminate conflicts between periodic perturbation and feature normalization.

CRediT authorship contribution statement

Yu Ran: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization; **Li Zhang:** Writing – review & editing, Validation, Resources, Investigation, Funding acquisition, Formal analysis; **Shi Yuan:** Validation, Resources, Investigation, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used chatGPT in order to improve the readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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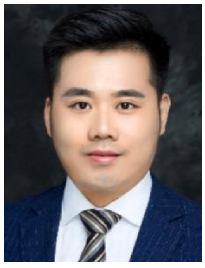
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